S1SBP6A

Bio-Processor

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Data Sheet

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Chip Handling Guide

Precaution against Electrostatic Discharge

When using semiconductor devices, ensure that the environment is protected against static electricity:

1. Wear antistatic clothes and use earth band.
2. All objects that are in direct contact with devices must be made up of materials that do not produce static electricity.
3. Ensure that the equipment and work table are earthed.
4. Use ionizer to remove electron charge.

Contamination

Do not use semiconductor products in an environment exposed to dust or dirt adhesion.

Temperature/Humidity

Semiconductor devices are sensitive to:

- Environment
- Temperature
- Humidity

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High temperature or humidity deteriorates the characteristics of semiconductor devices. Therefore, do not store or use semiconductor devices in such conditions.

Mechanical Shock

Do not to apply excessive mechanical shock or force on semiconductor devices.

Chemical

Do not expose semiconductor devices to chemicals because exposure to chemicals leads to reactions that deteriorate the characteristics of the devices.

Light Protection

In non- Epoxy Molding Compound (EMC) package, do not expose semiconductor IC to bright light. Exposure to bright light causes malfunctioning of the devices. However, a few special products that utilize light or with security functions are exempted from this guide.

Radioactive, Cosmic and X-ray

Radioactive substances, cosmic ray, or X-ray may influence semiconductor devices. These substances or rays may cause a soft error during a device operation. Therefore, ensure to shield the semiconductor devices under environment that may be exposed to radioactive substances, cosmic ray, or X-ray.

EMS (Electromagnetic Susceptibility)

Strong electromagnetic wave or magnetic field may affect the characteristic of semiconductor devices during the operation under insufficient PCB circuit design for Electromagnetic Susceptibility (EMS).

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List of Conventions

Register RW Access Type Conventions

Type	Definition	Description
R	Read Only	The application has permission to read the Register field. Writes to read-only fields have no effect.
W	Write Only	The application has permission to write in the Register field.
RW	Read & Write	The application has permission to read and writes in the Register field. The application sets this field by writing 1'b1 and clears it by writing 1'b0.

Register Value Conventions

Expression	Description
x	Undefined bit
X	Undefined multiple bits
?	Undefined, but depends on the device or pin status
Device dependent	The value depends on the device
Pin value	The value depends on the pin status

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Reset Value Conventions

Expression	Description
0	Clears the register field
1	Sets the register field
x	Don't care condition

Warning: Some bits of control registers are driven by hardware or write operation only. As a result the indicated reset value and the read value after reset might be different.

List of Acronyms

Acronyms	Descriptions
ADC	Analog to Digital Converter
AES	Advanced Encryption Standard
AFE	Analog Front End
AHB	Advanced High-performance Bus
APB	Advanced Peripheral Bus
BIA	Bioelectrical Impedance Analysis
BOR	Brown-Out Reset
CDM	Charged Device Model
CLK	Clock
CMRR	Common Mode Rejection Ratio
DAC	Digital to Analog Converter
DCC	DC Offset Cancellation
DC-DC	DC to DC Voltage Converter
DSP	Digital Signal Processing
DTRNG	Digital True Random Number Generator
DVFS	Dynamic Voltage and Frequency Scaling
ECG	Electrocardiography
ENOB	Effective Number Of Bit
fCLK	Modulator Clock
Fin	Input Frequency
FPU	Floating-point Unit
FRC	Free-running Counter
Fs	Sampling Frequency
FS	Full Scale
GPIO	General Purpose I/O
HBM	Human Body Model
HPF	High Pass Filter
I2C	Inter-Integrated Circuits
IA	Instrumentation Amplifier
IDC	Incremental Data Converter
LDO	Low-dropout Regulator
LPF	Low Pass Filter
MA	Motion Artifact
MCU	Micro Control Unit

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Acronyms	Descriptions
µDMAC	Micro Direct Memory Access Controller
MSPS	Mega Sample Per Second
OS	Operating System
OSC	Oscillator
PFM	Pulse Frequency Modulation
PGA	Programmable Gain Amplifier
POR	Power On Reset
PLL	Phase-locked Loop
PPG	Photoplethysmography
PWM	Pulse Width Modulation
RX	Receiver
RTC	Real-Time Clock
RTI	Referred to Input
SAR	Successive Approximation
SFR	Special Function Register
SNH	Sample-and-Hold
SPI	Serial Peripheral Interface
SPS	Sample Per Second
SRC	Sample Rate Converter
SRP	Samsung Reconfigurable Processor
Ta	Ambient Temperature
Tstg	Storage Temperature
THD	Total Harmonic Distortion
TIA	Trans impedance Amplifier
Tj	Junction Temperature
TMPX	Test Multiplexer
TX	Transmitter
UART	Universal Asynchronous Receiver and Transmitter
VCMI	Input Common Mode Voltage
VCO	Voltage Controlled Oscillator
WFE	Wait For Event
WFI	Wait For Interrupt
WLP	Wafer Level Package
XiP	eXecution-in-Place

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1 Product Overview

This chapter includes the following sections:

- Introduction
- Key features
- Block diagram

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Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

1.1 Introduction

S1SBP6A is a low-power Bio-Processor targeting multiple bio-sensor data acquisition and processing for healthcare and other applications. S1SBP6A supports a wide range of on-chip bio-sensor readouts such as Electrocardiography (ECG), Bioelectrical Impedance Analysis (BIA), optical (Photoplethysmography, PPG), and Galvanic Skin Response (GSR). In addition, General Purpose (GP) analog/digital interfaces allow acquiring data from external analog or digital sensors in a synchronized manner.

S1SBP6A integrates a Micro Controller Unit (MCU) powered by an ARM Cortex-M4 with Floating Point Unit (FPU) engine to improve the signal quality in real-time. And, Slim Samsung Reconfigurable Processor (SlimSRP) is utilized as a Digital Signal Processor (DSP) to improve the power efficiency of on-chip signal processing. In addition, S1SBP6A embeds 2 MB Flash, 256 KB SRAM for MCU, and 160 KB SRAM for DSP. Cortex-M4 core features the degrees of freedom of on-chip processing capability and DSP offers the efficient ways to implement various applications. Also, standard digital interfaces such as SPI, I2C, UART, and GPIO are supported for external communications.

Following are the features applied in the S1SBP6A:

- Clock gating
- Power gating with fine-grained power domains
- On-chip Flash Memory (2 MB)
- On-chip SRAM (256 KB for MCU and 160 KB for DSP)
- Dedicated low-power Analog Front Ends (AFEs) for ECG, PPG, BIA, and GSR applications
- Integrated LED drivers for PPG and analog current signal synthesizer for BIA
- Sample Rate Converter (SRC) as a hardware accelerator
- 12b GP analog readouts
- Real Time Clock (RTC) with system control
- Built-in Low-Dropout Regulators (LDOs) along with system POR and BOR
- Embedded PLL and Oscillators (OSC)
- Security IPs such as DTRNG (Digital True Random Number Generator) and AES (Advanced Encryption Standard)
- Versatile digital interfaces

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1.2 Key Features

S1SBP6A supports the following features:

- **Single channel ECG readout with built-in High Pass Filter (HPF)**
  - Low-noise of 3.5 μV_{rms} (0.5 Hz - 250 Hz)
  - High common-mode rejection capability, CMRR > 80dB
  - High impedance electrode interface
  - Maximum of 10 mV_{p-p} ECG inputs without significant signal distortion
  - Flexible gain and bandwidth control:
  - Gain of 100 V/V - 600 V/V, bandwidth of 75 Hz - 250 Hz
  - 12b, 1 kSPS/8 kSPS digital conversion
  - Built-in body bias driver capable of driving up to 10 μA current
  - Built-in HPF (< 0.5 Hz) to be tolerable to more than ±300 mV electrode offset
- **BIA readout with integrated current source**
  - Four points (tetra-polar electrode system) complex impedance measurements
  - 6b, 1 MSPS/4 MSPS sine-wave generation
  - Multi-frequency current excitation ranging from 1 kHz to 1 MHz
  - 12b, 1 kSPS/8 kSPS impedance conversion
  - Impedance calibration with external resistors
- **PPG readout with integrated LED drivers**
  - Built-in flexible LED driver with four independent current sink DACs
    - Maximum of four LEDs with different wavelengths (Green, Red, IR)
    - Flexible LED driving current level
    - 0 - 100 mA with 0.78 mA/step, or 0 - 50 mA with 0.39 mA/step
    - Flexible LED current pulse repetition
    - 100 Hz - 1 kHz LED pulse repetition rate with minimum 50 μs LED on time
  - Two synchronous Photo Diode (PD) readouts with ambient light sampling
    - Built-in DC level restoration with ±16 μA (6b) photocurrent cancellation
    - Wide dynamic range readout (10 kΩ - 2 MΩ) Low-noise (integrated over 0.5 Hz - 20 Hz)
    - Ambient light sampling synchronous to LED pulse
    - Maximum of four pulse slots synchronous signal sampling for each readout front end
    - 24b, 1 kSPS/pulse digitization
- **GSR readouts with integrated voltage sources**
  - Two points (bipolar electrode system) conductance measurements
  - Built-in DC voltage source with DC/AC-mode (4 Hz, Typ.) outputs
  - Wide dynamic range conductance measurement (0.05 μS - 100 μS)
  - 24b, 500 SPS conductance conversion

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- **GP analog interfaces**
  - 12b, 1 kSPS/8 kSPS GP analog readouts
    - Voltage Gain Amplifier (VGA), Trans-Impedance Amplifier (TIA), Low Pass Filter (LPF) input interfaces
    - Wide range of parameter programmability
    - Synchronous data acquisition to the other AFEs
    - Digital data decimation
- **Analog peripherals**
  - Four integrated LDOs (1.1 V) to supply AFE, DSP/MCU, memory, and clock domain
  - Built-in Power On Reset (POR), and Brown-Out Reset (BOR)
  - Built-in PLL (102.4 MHz) and oscillators (4.096 MHz, 24.576 MHz)
  - Built-in bandgap reference and junction temperature sensor
- **Micro controller unit**
  - ARM Cortex-M4 with FPU with maximum 102.4 MHz operation
  - 2 MB embedded Flash and 256 KB SRAM
  - SlimSRP with 160 KB SRAM which is shared by MCU
  - Sample Rate Converter (SRC) as a hardware accelerator
  - 32 channel Micro Direct Memory Access Controller (µDMAC)
  - Seven Timers, one Pulse Width Modulation (PWM) Timer, one Dual-Input Timer, one Watchdog, and one Real-Time Clock (RTC)
  - Serial Flash Interface with eXecution-in-Place (XiP) capability
  - JTAG/SW interface for debug
  - Clock Management Unit (CMU) and Power Management Unit (PMU)
  - Digital True Random Number Generator (DTRNG) and Advanced Encryption Standard (AES)
- **Digital interfaces**
  - Five Inter-Integrated Circuits (I2C)
  - Five Serial Peripheral Interfaces (SPI)
  - Three Universal Asynchronous Receiver and Transmitters (UART)
  - 48 General Purpose I/Os (GPIO)
- **Package**
  - 154 pin WLP 4.76 × 4.66 × 0.43 mm, 0.35 mm pin pitch

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1.3 Block Diagram

Figure 1 illustrates the block diagram of S1SBP6A.

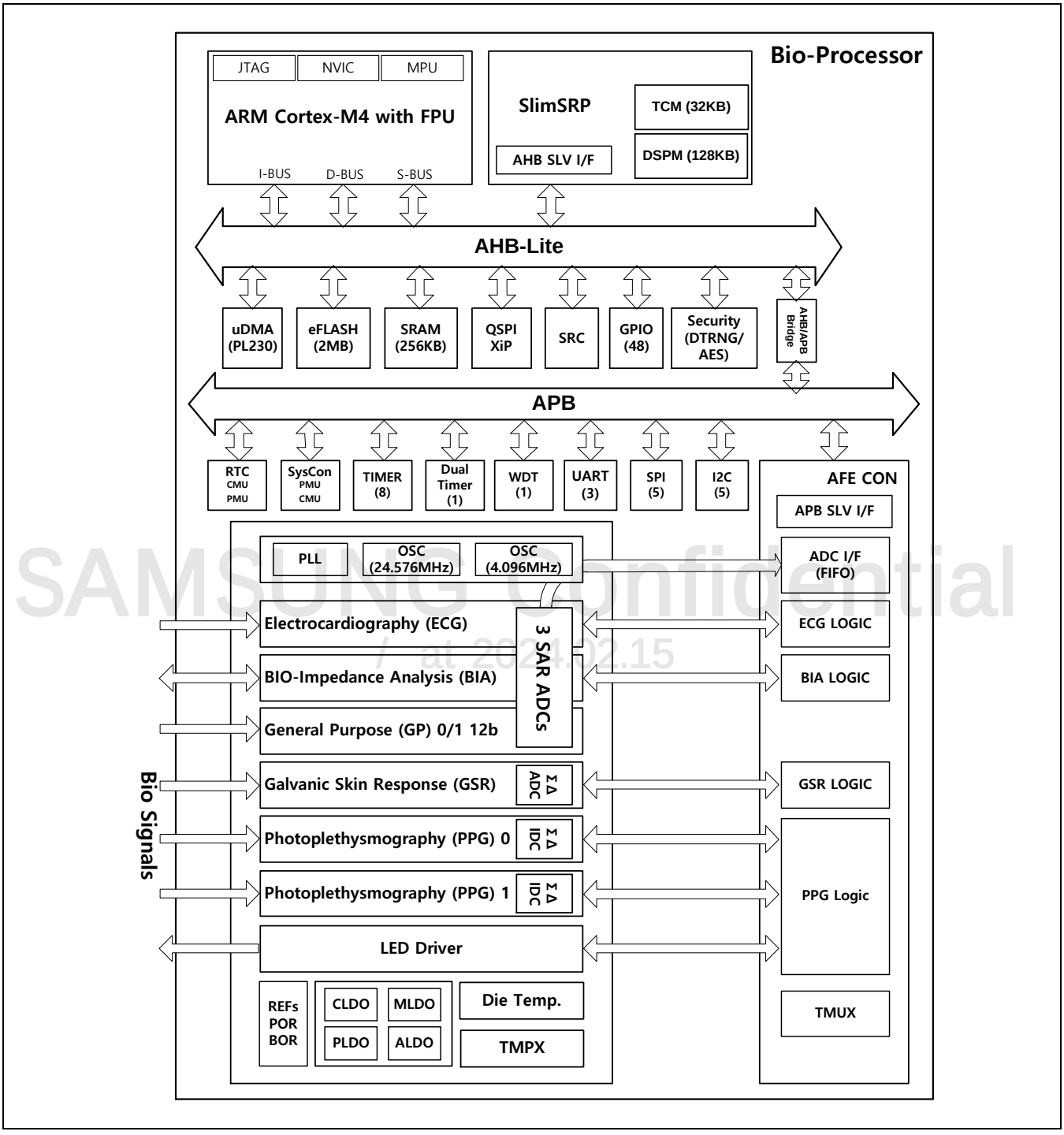


Figure 1 Block Diagram of S1SBP6A

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2 Operating Conditions and Electrical Characteristics

This chapter includes the following sections:

- Overview
- Absolute maximum ratings
- Recommended operating conditions
- Electrical characteristics of MCU
- Electrical characteristics of AFE
- Electrical characteristics of PMIPs

2.1 Overview

This chapter describes the operating conditions and electrical specifications of digital blocks and Analog Front Ends implemented in S1SBP6A. These specifications are preliminary and results of simulation at 25 °C. These are subject to change.

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2.2 Absolute Maximum Ratings

Table 1 describes the absolute maximum ratings.

Table 1 Absolute Maximum Ratings

Parameter	Value	Unit
Analog Power to Analog Ground	–0.3 to 3.3	V
Analog Ground to Digital Ground	–0.3 to 0.3	V
Electrostatic Discharge Ratings: Human Body Model (HBM)	–7.0 to 7.0	kV
Electrostatic Discharge Ratings: Charged Device Model (CDM)	–500 to 500	V
Storage Temperature: Tstg	TBD	°C

NOTE: These ratings are associated only with stress. It does not imply any functional operation of the condition described in Section *2.2 Absolute Maximum Ratings*. Exposure to the absolute maximum rated conditions for long duration affects the reliability of the device.

2.3 Recommended Operating Conditions

Table 2 describes the recommended operating conditions.

Table 2 Recommended Operating Conditions

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Parameter		Min.	Typ.	Max.	Unit
Analog Supply Voltage of S1SBP6A	AVDD33A	2.7	3	3.3	V
Analog Supply Voltage for Built-in LDOs	AVDD33	2.7	3	3.3	V
LED Driver Supply Voltage	AVDD33_LED	2.7	3	3.3	V
Analog Ground	AVSS33	–	0	–	V
LED Driver Ground	AVSS33_LED	–	0	–	V
Digital Supply Voltage	DVDD33	1.65	3	3.6	V
Digital Ground	DVSS33	–	0	–	V
Operating Ambient Temperature	Ta	–25	–	80	°C

NOTE: External noise filtering is recommended to supply voltage through AVDD33A.

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2.4 Electrical Characteristics of MCU

Table 3 describes the electrical characteristics of MCU.

Table 3 Electrical Characteristics of MCU

Parameters		Conditions	Min.	Typ.	Max.	Unit
Clock Frequency	AHB clock	When PLL is used.	–	–	102.4	MHz
	APB clock	When PLL is used.	–	–	102.4	MHz
	DSP clock	When PLL is used.	–	–	70	MHz
	AFE control clock	When one of External 4.096 MHz crystal input, HSOSC, and LSOSC is used.	–	4.096	–	MHz
	RTC clock	External 32.768 kHz crystal input	–	32.768	–	kHz
Current Consumption	LPM0	For more information, refer to *Table 15*	–	25	–	µA
	LPM1	For more information, refer to *Table 15*	–	TBD	–	µA
	LPM2	For more information, refer to *Table 15*	–	TBD	–	µA
	LPM3	For more information, refer to *Table 15*	–	TBD	–	µA
	LPM4	For more information, refer to *Table 15*	–	TBD	–	µA
	ACTIVE MCU	CoreMark, AHB clock = 102.4 MHz	–	TBD	–	mA
		CoreMark, AHB clock = 51.2 MHz	–	TBD	–	mA
		CoreMark, AHB clock = 25.6 MHz	–	TBD	–	mA
	ACTIVE DSP	CoreMark, SRP clock = 70 MHz	–	TBD	–	mA
		CoreMark, SRP clock = 51.2 MHz	–	TBD	–	mA
		CoreMark, SRP clock = 25.6 MHz	–	TBD	–	mA
Wakeup Time		Wake up from LPM0	–	TBD	–	µs
		Wake up from LPM1	–	TBD	–	µs
		Wake up from LPM2	–	TBD	–	µs
		Wake up from LPM3	–	TBD	–	µs
		Wake up from LPM4	–	TBD	–	µs

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2.5 Electrical Characteristics of AFE

2.5.1 ECG

Table 4 describes the electrical characteristics of ECG.

Table 4 Electrical Characteristics of ECG

Parameter	Condition	Min.	Typ.	Max.	Unit
Max. Differential Input	–	–	10	–	mV_{p-p}
Common-Mode Input	–	–	0.55	–	V
Max. Differential DC Offset	–	0	–	AVDD 33A	V
Gain of IA	–	–	100	–	V/V
Gain of PGA	Gain variation (± 20 %)	1	–	6	V/V
Total Channel Gain	Gain of IA × Gain of PGA	100	–	600	V/V
Bandwidth (LPF)	–	75	–	250	Hz
Bandwidth (HPF)	–	–	0.5	–	Hz
Input Impedance	Measured at 60 Hz	–	100	–	MΩ
CMRR @ 60 Hz	Without input source mismatch	–	80	–	dB
Input Noise	Measured at 0.5 - 250 Hz	–	3.5	–	μV_{rms}
ADC Resolution	–	–	12	–	bit
ADC Sampling Speed	–	1	–	8	kHz
ADC Reference Voltage	–	–	1.1	–	V
Total Harmonic Distortion (THD)	Measured at 10 mV_{p-p} input	–	–	1	%
Body Bias Drivability	–	–	10	–	μA
Power Consumption	Inc. body bias voltage source Exc. LDO power consumption	–	42	–	μW

NOTE: The performances are not including non-ideal effects of electrodes.

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2.5.2 PPG

Table 5 describes the electrical characteristics of PPG.

Table 5 Electrical Characteristics of PPG

Parameter		Condition	Min.	Typ.	Max.	Unit
LED Driver	# of LED Driver	2 for RED/IR 2 for GREEN	–	4	–	ea
	LED Driver Output Voltage	–	1.2	–	–	V
	Operation Current	–	–	523	–	μA
	LED Driving Current Range	0.78 mA/step, Max current 0.39 mA/step, Max current	–	100 50	–	mA mA
	LED Driving Current Resolution	–	–	7	–	bit
	LED Pulse Repetition Rate	–	100	–	1000	Hz
	LED On Time	–	50	–	–	μs
PPG Receiver	# of Photo Diode (PD)	–	–	2	–	ea
	PD Capacitance	–	–	–	100	pF
	DCC DAC Current	–	-16	–	16	μA
	DCC Resolution	–	–	6	–	bit
	Trans-Impedance	–	10	–	2000	kΩ
	Input Noise	Inc. DCC (16 μA output) Inc. DCC (8 μA output) Exc. DCC (Rf = 250 kΩ, Cf = 6 pF)	– –	200 100 30	–	pA_{rms}
	Receiver SNR	BW: 20 Hz TIA: 500 kΩ 50 % FS output, LED pulse repetition: 1 kHz Exc. DC cancellation	–	90	–	dBFS
	Data Rate	–	100	–	1000	Hz
	ADC Resolution	–	–	24	–	bit
	ADC Sampling speed	–	–	2.048	–	MHz
	Power Consumption	Pulse repetition:1 kSPS LED on time: 100 μs, Per 1ch. 10 % active power control For more information, refer to Figure 14	–	600	–	μW

NOTE: The performances are not including non-ideal effects of LEDs and PDs.

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2.5.3 BIA

Table 6 describes the electrical characteristics of BIA.

Table 6 Electrical Characteristics of BIA

Parameter		Condition	Min.	Typ.	Max.	Unit
TX	DAC Resolution	–	–	6	–	bit
	DAC FS Voltage	Common-mode voltage = 0.55 V	0.3	0.5	0.6	V_{p-p}
	Driving Current	Ext. resistor = 0 kΩ Int. resistor=1.67 kΩ	–	105	–	μA_{rms}
		Ext. resistor = 4.3 kΩ Int. resistor=0.67 kΩ DAC FS Voltage=0.4 V_{p-p}	–	28	–	μA_{rms}
	DAC Sampling Speed	–	–	–	4	MHz
	Filter Bandwidth	–	125	–	2000	kHz
	Sine-Wave Frequency	–	1	–	1000	kHz
	Operation Current	Driving current 105 μA_{rms}	–	700	–	μA
RX	Impedance Measurement Range	VDD = 1.1 V	–	TBD	–	kΩ
	Gain	PGA	0.5	–	12	V/V
	Bandwidth	Int. resistor = 5 kΩ Ext. capacitor = 4.7 μF	–	6.7	–	Hz
	Input Impedance	–	–	500	–	kΩ
	Nonlinearity	500 Ω - 2 kΩ	–	1	–	%
	Operation Current	–	–	400	–	μA
	ADC Resolution	–	–	12	–	bit
	ADC Sampling Speed	–	1	–	8	kHz
	ADC Reference Voltage	–	–	1.1	–	V

NOTE: The performances are not including non-ideal effects of electrodes.

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2.5.4 GSR

Table 7 describes the electrical characteristics of GSR.

Table 7 Electrical Characteristics of GSR

Parameter		Condition	Min.	Typ.	Max.	Unit
GSR	Output Voltage Range	–	0	–	2.5	V_{p-p}
	Pulse Frequency	4 Hz default	–	4	–	Hz
	Linearity	10 kΩ - 20 MΩ	–	3	–	%
	Operation Current	–	–	248	–	μA
	GSR Measurement Range	–	0.05	–	100	μS
	ADC Resolution	–	–	24	–	bit
	ADC Data Rate	–	125	500	1000	SPS
	Effective Number of Bit (ENOB)	Fs = 500 Hz Fin = 240 Hz	–	13.5	–	Bit
	Operation Clock	–	–	64	–	kHz
	ADC Reference Voltage	–	–	2.5	–	V

NOTE: The performances are not including non-ideal effects of electrodes.

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2.5.5 GP Analog Interfaces

2.5.5.1 12b GP Analog Readouts

Table 8 describes the analog readouts of 12b GP.

Table 8 Electrical Characteristics of 12b GP Analog Readouts

Parameter		Condition	Min.	Typ.	Max.	Unit
Trans-Impedance Amplifier(TIA)	Trans-Impedance (TI)	For more information, refer to *Table 20*.	0.6	–	10.4	MΩ
	Bandwidth		1	–	35.3	kHz
	Input Current Range	1.4 V_{p-p}/TI	0.1	–	1.65	μA_{rms}
	Signal-to-Noise Ratio	Input Current = 0.125 μA, TI = 10.4 MΩ	–	58	–	dB
Variable Gain Amplifier(VGA)	Gain	For more information, refer to *Table 21*.	2	–	16	V/V
	Bandwidth		1	–	2	kHz
	Input Voltage Range	1.4 V_{p-p}/Gain	–	–	700	mV_{p-p}
	Signal-to-Noise Ratio	Input = 0.04 V_{p-p}, Gain = 16	–	56	–	dB
Low Pass Filter (LPF)	Bandwidth	Differential-mode For more information, refer to *Table 22*	1	–	16.3	kHz
Voltage Attenuator	Gain (NOTE)	For more information, refer to *Figure 31*	–	0.2	–	V/V
	Int. Attenuator Resistor	–	–	100	–	kΩ
	Input Voltage Range	–	–	–	1.4	V_{p-p}
Die Temp.	Temperature Range		10	–	85	°C
Total Harmonic Distortion (THD)		-3 dB FS, GP Analog Full Path	–	–	1	%
ADC Reference		–	–	1.1	–	V
ADC Speed		–	1	–	8	kHz
ADC Resolution		–	–	12	–	bit
Signal-to-Noise Ratio		Fs = 8 kSPS Fin = 137 Hz	–	65	–	dB
Signal-to-Noise and Distortion Ratio		Fs = 8 kSPS Fin = 137 Hz	–	64	–	dB
Spurious-Free Dynamic Range		Fs = 8 kSPS Fin = 137 Hz	–	71.2	–	dB
Operation Current		GP Analog + ADC	–	42	–	μA

NOTE: Rext = 200 kΩ Gain is determined depending on the value of Rext.

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2.6 Electrical Characteristics of PMIPs and Clock Generators

2.6.1 PMIP

Table 9 describes the electrical characteristics of PMIPs.

Table 9 Electrical Characteristics of PMIPs

Parameter		Condition	Min.	Typ.	Max.	Unit
Input Voltage		–	2.7	–	3.3	V
Output Voltage	ALDO	–	–	1.1	–	V
	PLDO	–	–	1.1	–	V
	MLDO	–	–	1.1	–	V
	CLDO	–	–	1.1	–	V
Max. Load Current	ALDO	–	–	20	–	mA
	PLDO	–	–	10	–	mA
	MLDO	–	–	30	–	mA
	CLDO	–	–	40	–	mA
Line Regulation	ALDO	–	–	20	–	mV/V
	PLDO	–	–	25	–	mV/V
	MLDO	–	–	10	–	mV/V
	CLDO	–	–	10	–	mV/V
Load Regulation	ALDO	–	–	20	–	mV/A
	PLDO	–	–	20	–	mV/A
	MLDO	–	–	10	–	mV/A
	CLDO	–	–	10	–	mV/A
Standby Current Consumption	ALDO	–	–	4	–	µA
	PLDO	–	–	2	–	µA
	MLDO	–	–	7	–	µA
	CLDO	–	–	7	–	µA
Irush Current	ALDO	–	–	TBD	–	A
	PLDO	–	–	TBD	–	A
	MLDO	–	–	TBD	–	A
	CLDO	–	–	TBD	–	A
Start-up Time	ALDO	–	–	0.1	–	ms
	PLDO	–	–	0.1	–	ms
	MLDO	–	–	0.1	–	ms
	CLDO	–	–	0.1	–	ms
Threshold Voltage	POR	–	1.52	–	1.62	V

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Parameter		Condition	Min.	Typ.	Max.	Unit
	BOR	–	2.6	–	2.7	V
	Operation Current	–	–	7.5	–	μA

2.6.2 Clock Generators

- **Phase-Locked Loop**

Table 10 describes the electrical characteristics of Phase-Locked Loop.

Table 10 Electrical Characteristics of Phase-Locked Loop

Parameter	Condition	Min.	Typ.	Max.	Unit
Operating Current	PLDO_OUT = 1.1 V, 50 MHz, 25 °C	–	1.5	–	mA
Power Down Current	RESETB = 0, 25 °C	–	0.5	–	μA
Input Frequency	–	4	16	100	MHz
Reference Frequency	–	2	4	10	MHz
PLL Output Frequency	4.096 MHz Crystal Input	–	102.4	–	MHz
VCO Output Frequency	–	100	–	200	MHz
Duty Ratio	–	40	–	60	%
Lock Time (1)	–	–	–	200	cycles
Period Jitter	–	–	–	± 1.5 (2)	%
Frequency Setting Range	–	0.71	–	1.64	MHz

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NOTE:

1 1 cycle = 1/(Reference Frequency) = 1/(Input Frequency/pre-divider value)

2 ± 1.5 of VCO period

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- **High Speed Oscillator**

Table 11 describes the electrical characteristics of High Speed Oscillator.

Table 11 Electrical Characteristics of High Speed Oscillator

Parameter	Condition	Min.	Typ.	Max.	Unit
Operating Current	PLDO_OUT = 1.1 V, 25 °C	–	12	–	μA
Power Down Current	RESETB = 0, 25 °C	–	–	–	μA
Short-term Period Jitter	–	–	–	15 %	Period
Output Frequency by CS (NOTE)	Under all PVT variation	–2 %	24.576	+2 %	MHz
Duty Ratio by CS (NOTE)	–	20	50	80	%
Lock Time by CS (NOTE)	–	–	–	300	μs

NOTE: CS means calibration scheme. Oscillator frequency can be calibrated using external 32.768 kHz clock.

- **Low Speed Oscillator**

Table 12 describes the electrical characteristics of **Low-Speed Oscillator.**

Table 12 Electrical Characteristics of Low-Speed Oscillator

Parameter	Condition	Min.	Typ.	Max.	Unit
Operating Current	PLDO_OUT = 1.1 V, 25 °C	–	8	–	μA
Power Down Current	RESETB = 0, 25 °C	–	–	–	μA
Short-term Period Jitter	–	–	–	15 %	Period
Output Frequency by CS (NOTE)	Under all PVT variation	–2 %	4.096	+2 %	MHz
Duty Ratio by CS (NOTE)	–	20	50	80	%
Lock Time by CS (NOTE)	–	–	–	300	μs

NOTE: CS means calibration scheme. Oscillator frequency can be calibrated using external 32.768 kHz clock

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3 Ball Map and Pin Description

This chapter includes the following sections:

- Overview
- Ball map
- Pin description

3.1 Overview

This chapter describes S1SBP6A ball map and signal pins.

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3.2 Ball Map

Figure 2 illustrates the ball map of S1SBP6A device.

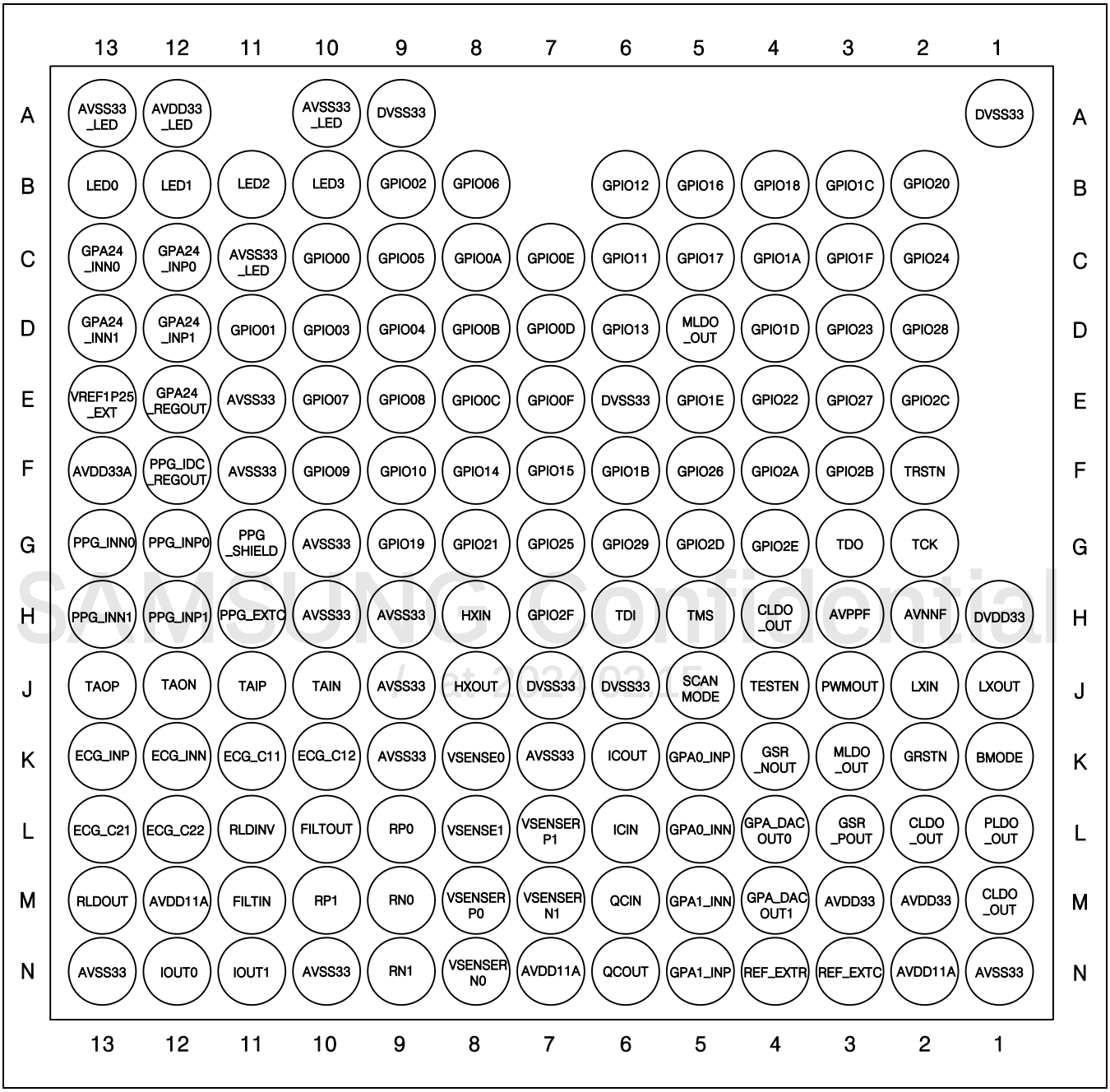


Figure 2 Ball Map of S1SBP6A

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3.3 Pin Description

Table 13 describes the pin assignment.

Table 13 Pin Description of S1SBP6A

No.	Ball	PIN	Description	I/O
1	B12	LED1	LED Driver1 Output (tolerable up to 5 V)	AIO
2	B13	LED0	LED Driver0 Output (tolerable up to 5 V)	AIO
3	C12	GPA24_INP0	Positive Analog Signal Input for GSR (24b)	AI
4	C13	GPA24_INN0	Negative Analog Signal Input for GSR (24b)	AI
5	D12	GPA24_INP1	No use	AI
6	D13	GPA24_INN1	No use	NC
7	C11	AVSS33	Analog Ground	AG
8	E13	VREF1P25_EXT	External 1.25 V Reference Voltage Input	AI
9	E12	GPA24_REGOUT	GSR (24b) Regulator Output	AO
10	F12	PPG_IDC_REGOUT	IDC (24b) Regulator Output	AO
11	F13	AVDD33A	3 V Analog Power Supply	AP
12	G12	PPG_INP0	Photo Diode Cathode Input0	AI
13	G13	PPG_INN0	Photo Diode Anode Input0	AI
14	G11	PPG_SHIELD	Photo Diode Shied Voltage Output	AO
15	H11	PPG_EXTC	External Capacitor for DCC Noise Filtering	AI
16	H13	PPG_INN1	Photo Diode Anode Input0	AI
17	H12	PPG_INP1	Photo Diode Cathode Input0	AI
18	J11	TAIP	External Positive Input for Analog Buffer to Monitor Internal Analog Signal	AI
19	J13	TAOP	Positive Analog Buffer Output	AO
20	J12	TAON	Negative Analog Buffer Output	AO
21	J10	TAIN	External Negative Input for Analog Buffer to Monitor Internal Analog Signal	AI
22	K12	ECG_INN	Negative ECG Signal Input	AI
23	K13	ECG_INP	Positive ECG Signal Input	AI
24	K11	ECG_C11	External Capacitor Port1 for Negative Input of HPF2 in ECG Readout	AIO
25	K10	ECG_C12	External Capacitor Port2 for Negative Input of HPF2 in ECG Readout	AIO
26	L13	ECG_C21	External Capacitor Port1 for Positive Input of HPF2 in ECG Readout	AIO
27	L12	ECG_C22	External Capacitor Port2 for Positive Input of HPF2 in ECG Readout	AIO
28	L11	RLDINV	Negative Input of ECG Electrode Driver for Body Bias	AI
29	M13	RLDOUT	Output of ECG Electrode Driver for Body Bias	AO
30	M12	AVDD11A	ALDO output (1.1 V) for ECG, BIA, GPA0/1, 12b ADC, 10b GP DAC	AO
31	L10	FILTOUT	BIA DAC Filter Output	AO
32	M11	FILTIN	BIA Current Converter Input	AI

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No.	Ball	PIN	Description	I/O
33	N12	IOUT0	BIA Current Driver Output0	AO
34	N11	IOUT1	BIA Current Driver Output1	AO
35	L9	RP0	BIA Calibration Resistor0 Positive Port	AIO
36	M10	RP1	BIA Calibration Resistor1 Positive Port	AIO
37	M9	RN0	BIA Calibration Resistor0 Negative Port	AIO
38	N9	RN1	BIA Calibration Resistor1 Negative Port	AIO
39	N10	AVSS33	Analog Ground	AG
40	K8	VSENSE0	BIA Differential Amplifier Input0 for Voltage Sensing	AI
41	L8	VSENSE1	BIA Differential Amplifier Input1 for Voltage Sensing	AI
42	M8	VSENSERP0	BIA Differential Amplifier Input0 for Voltage Sensing from Resistor0	AI
43	L7	VSENSERP1	BIA Differential Amplifier Input0 for Voltage Sensing from Resistor1	AI
44	N8	VSENSERN0	BIA Differential Amplifier Input1 for Voltage Sensing from Resistor0	AI
45	M7	VSENSERN1	BIA Differential Amplifier Input1 for Voltage Sensing from Resistor1	AI
46	N7	AVDD11A	ALDO output (1.1 V) for ECG, BIA, GPA0/1, 12b ADC, 10b GP DAC	AO
47	K6	ICOUT	I channel demodulator low pass filter, connect 4.7 μF between ICOUT and Ground	AIO
48	L6	ICIN	I channel demodulator low pass filter, connect 4.7 μF between ICIN and Ground	AI
49	N6	QCOUT	Q channel demodulator low pass filter, connect 4.7 μF between QCOUT and Ground	AIO
50	M6	QCIN	Q channel demodulator low pass filter, connect 4.7 μF between QCIN and Ground	AI
51	K5	GPA0_INP	GP Analog0 Positive Analog Signal Input	AI
52	L5	GPA0_INN	GP Analog0 Negative Analog Signal Input	AI
53	N5	GPA1_INP	GP Analog1 Positive Analog Signal Input	AI
54	M5	GPA1_INN	GP Analog1 Negative Analog Signal Input	AI
55	L4	GPA_DACOUT0	No use	AO
56	M4	GPA_DACOUT1	No use	AO
57	K4	GSR_NOUT	Negative GSR Electrode Connecting Pin	AIO
58	L3	GSR_POUT	Positive GSR Electrode Connecting Pin	AIO
59	N4	REF_EXTR	External Resistor (75 kΩ) for On-Chip Bias Generation	AIO
60	N3	REF_EXTC	External Noise Filtering Capacitor (4.7 μF) for On-Chip 1.25 V Reference	AIO
61	N2	AVDD11A	ALDO output (1.1 V) for ECG, BIA, GPA0/1, 12b ADC, 10b GP DAC	AO
62	M3	AVDD33	3 V Analog Power Supply for LDOs	AP
63	N1	AVSS33	Analog Ground	AG
64	M2	AVDD33	3 V Analog Power Supply for LDOs	AG

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No.	Ball	PIN	Description	I/O
65	M1	CLDO_OUT	LDO Output for MCU and Slim-SRP	AO
66	L2	CLDO_OUT	LDO Output for MCU and Slim-SRP	AO
67	K3	MLDO_OUT	LDO Output for SRAM and Flash Memory	AO
68	L1	PLDO_OUT	LDO Output for PLL and Oscillators	AO
69	K2	GRSTN	Global Reset	DIO
70	K1	BMODE	At Start Up (0:Embedded Flash Boot Mode, 1: External Serial, Flash Boot Mode), After Start Up (Wakeup Source)	DIO
71	J5	SCANMODE	Scan Mode Enable	DIO
72	J4	TESTEN	Test Mode Enable	DIO
73	J3	PWMOUT	PWM Out	DIO
74	J1	LXOUT	32.768 kHz Crystal Output	DIO
75	J2	LXIN	32.768 kHz Crystal Input	DIO
76	J6	DVSS33	Digital Ground	DG
77	H1	DVDD33	Digital Power	DP
78	J8	HXOUT	4.096 MHz Crystal Output	DIO
79	H8	HXIN	4.096 MHz Crystal Input	DIO
80	H2	AVNNF	Negative High Voltage Forcing and Monitoring Port (–9 V ~ 0 V)	AIO
81	H3	AVPPF	Positive High Voltage Forcing and Monitoring Port (0 V ~ 12 V)	AIO
82	H4	CLDO_OUT	LDO Output for MCU and Slim-SRP	AO
83	J7	DVSS33	Digital Ground	DG
84	G2	TCK	Test Clock	DIO
85	H5	TMS	Test Mode Selection	DIO
86	H6	TDI	Test Data Input	DIO
87	G3	TDO	Test Data Output	DIO
88	F2	TRSTN	TRSTN(Active Low)	DIO
89	H7	GPIO2F	GPIO2F/QSPI_CLK	DIO
90	G4	GPIO2E	GPIO2E/QSPI_CS	DIO
91	G5	GPIO2D	GPIO2D/QSPI_SI	DIO
92	E2	GPIO2C	GPIO2C/QSPI_SO	DIO
93	F3	GPIO2B	GPIO2B/QSPI_WP	DIO
94	F4	GPIO2A	GPIO2A/Timer0_EXT/QSPI_HLD	DIO
95	G6	GPIO29	GPIO29/I2C0_SDA	DIO
96	D2	GPIO28	GPIO28/I2C0_SCL	DIO
97	E3	GPIO27	GPIO27/UART0_RTSN	DIO
98	F5	GPIO26	GPIO26/UART0_CTSN	DIO
99	G7	GPIO25	GPIO25/UART0_RX	DIO

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No.	Ball	PIN	Description	I/O
100	C2	GPIO24	GPIO24/UART0_TX	DIO
101	D3	GPIO23	GPIO23/SPI0_MISO	DIO
102	E4	GPIO22	GPIO22/SPI0_MOSI	DIO
103	A1	DVSS33	Digital Ground	DG
104	G8	GPIO21	GPIO21/SPI0_CSN	DIO
105	B2	GPIO20	GPIO20/SPI0_CLK	DIO
106	C3	GPIO1F	GPIO1F/I2C4_SDA	DIO
107	E5	GPIO1E	GPIO1E/I2C4_SCL	DIO
108	D4	GPIO1D	GPIO1D/I2C3_SDA	DIO
109	B3	GPIO1C	GPIO1C/I2C3_SCL	DIO
110	F6	GPIO1B	GPIO1B/I2C2_SDA	DIO
111	C4	GPIO1A	GPIO1A/I2C2_SCL	DIO
112	G9	GPIO19	GPIO19/I2C1_SDA	DIO
113	B4	GPIO18	GPIO18/I2C1_SCL	DIO
114	D5	MLDO_OUT	LDO Output for SRAM and Flash Memory	AO
115	E6	DVSS33	Digital Ground	DG
116	C5	GPIO17	GPIO17/UART2_RTSN/PWMOUT	DIO
117	B5	GPIO16	GPIO16/UART2_CTSN/32.768 kHz	DIO
118	F7	GPIO15	GPIO15/UART2_Rx	DIO
119	F8	GPIO14	GPIO14/UART2_Tx	DIO
120	D6	GPIO13	GPIO13/UART1_RTSN/PTimer_EXT	DIO
121	B6	GPIO12	GPIO12/UART1_CTSN/Timer6_EXT	DIO
122	C6	GPIO11	GPIO11/UART1_Rx	DIO
123	F9	GPIO10	GPIO10/UART1_Tx/Timer5_EXT	DIO
124	E7	GPIO0F	GPIO0F/SPI4_MISO/Timer4_EXT	DIO
125	C7	GPIO0E	GPIO0E/SPI4_MOSI/Timer3_EXT	DIO
126	D7	GPIO0D	GPIO0D/SPI4_CSN/Timer2_EXT	DIO
127	E8	GPIO0C	GPIO0C/SPI4_CLK/Timer1_EXT	DIO
128	D8	GPIO0B	GPIO0B/SPI3_MISO	DIO
129	C8	GPIO0A	GPIO0A/SPI3_MOSI	DIO
130	F10	GPIO09	GPIO09/SPI3_CSN	DIO
131	E9	GPIO08	GPIO08/SPI3_CLK	DIO
132	E10	GPIO07	GPIO07/SPI2_MISO	DIO
133	B8	GPIO06	GPIO06/SPI2_MOSI	DIO
134	C9	GPIO05	GPIO05/SPI2_CSN	DIO

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No.	Ball	PIN	Description	I/O
135	D9	GPIO04	GPIO04/SPI2_CLK	DIO
136	D10	GPIO03	GPIO03/SPI1_MISO	DIO
137	B9	GPIO02	GPIO02/SPI1_MOSI	DIO
138	D11	GPIO01	GPIO01/SPI1_CSN	DIO
139	C10	GPIO00	GPIO00/SPI1_CLK	DIO
140	A9	DVSS33	Digital Ground	DG
141	A10	AVSS33_LED	Ground for LED Driver	AG
142	B10	LED3	LED Driver3 Output (tolerable up to 3 V)	AIO
143	B11	LED2	LED Driver2 Output (tolerable up to 3 V)	AIO
144	A13	AVSS33_LED	Ground for LED Driver	AG
145	A12	AVDD33_LED	3 V Analog Power Supply for LED Drivers	AP
146	E11	AVSS33	Analog Ground	AG
147	F11	AVSS33	Analog Ground	AG
148	G10	AVSS33	Analog Ground	AG
149	H10	AVSS33	Analog Ground	AG
150	H9	AVSS33	Analog Ground	AG
151	J9	AVSS33	Analog Ground	AG
152	K9	AVSS33	Analog Ground	AG
153	K7	AVSS33	Analog Ground	AG
154	N13	AVSS33	Analog Ground	AG

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NOTE: AI: Analog Input
DI: Digital Input
AO: Analog Output
DO: Digital Output
AIO: Analog Input/Output
AP: Analog Power
AG: Analog Ground
DIO: Digital Input/Output
DP: Digital Power
DG: Digital Ground
P: Power
G: Ground
NC: No Connection

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4 MCU

This chapter includes the following sections:

- Overview
- Subsystems
- System control

4.1 Overview

S1SBP6A integrates MCU, DSP, and AFE circuit controllers. MCU takes master control role of S1SBP6A such as managing Operating System (OS), signal processing, interfacing with external devices, and other common tasks. MCU comprises ARM Cortex-M4 core as a main processor, 2 MB embedded Flash with 2 KB Cache, 256 KB SRAM, Security IPs, and various peripherals.

Slim Samsung Reconfigurable Processor (SlimSRP) as a DSP of S1SBP6A is a power-efficient processor which can be used from low to middle complexity applications. It utilizes a reconfigurable processing core consisting four functional units. Cortex-M4 core controls SlimSRP as a slave subsystem through AHB-Lite bus interface. AHB masters such as Cortex-M4 core, DMAC, and SRC can utilize the SRAM inside SlimSRP.

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Following are the features available on S1SBP6A:

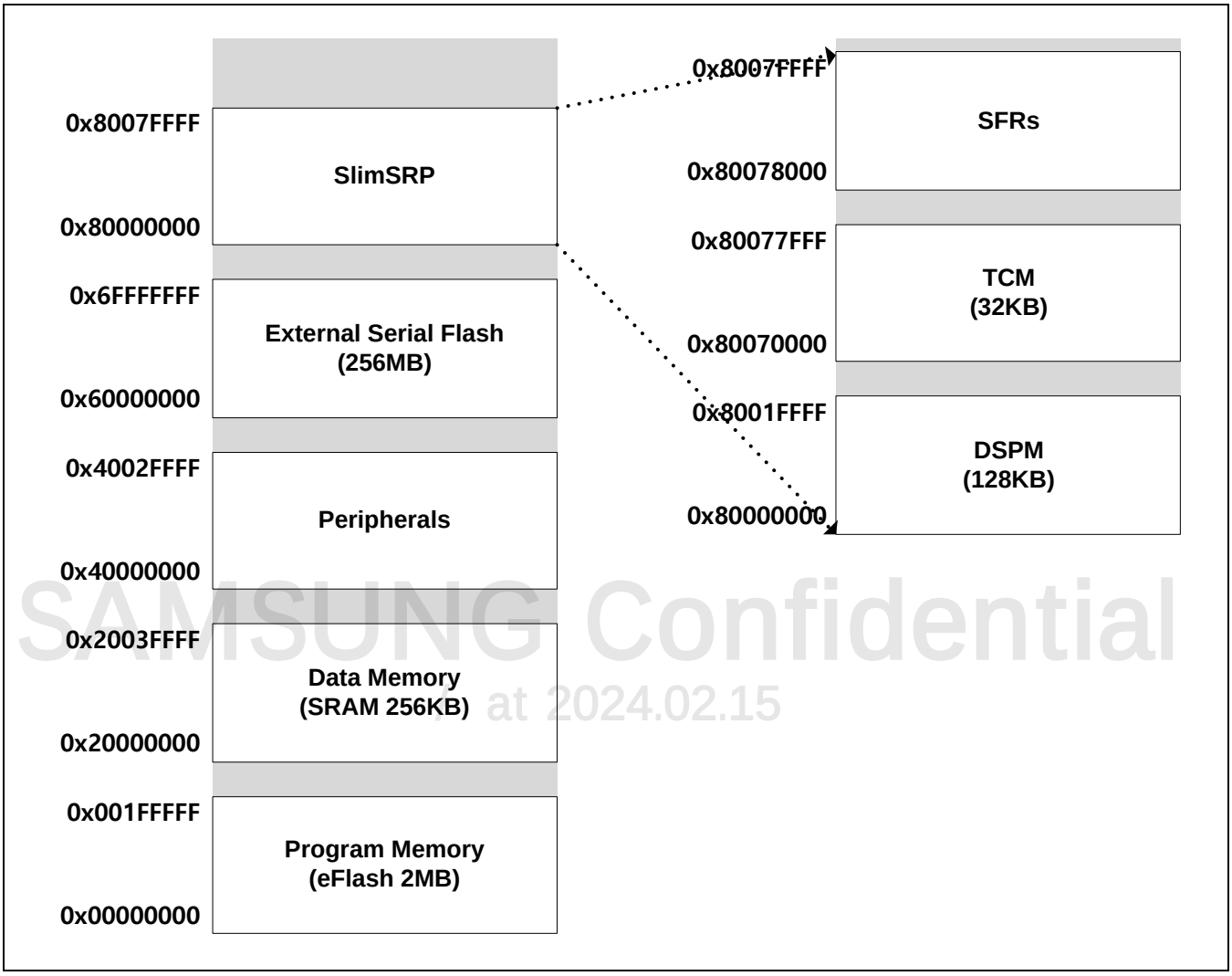
- MCU with Cortex-M4 core with FPU
- 2 MB embedded Flash with 2 KB Cache and 256 KB SRAM
- System control such as RTC, CMU, and PMU
- 32 channel μDMAC
- Three channel UARTs, five channel SPIs, five channel I2Cs, and 48 Pin GPIO (16 pin GPIO 3 ports)
- Seven Timers, one PWM Timer, one Dual-input Timer, one Watchdog, and one RTC
- Security IPs such as AES and DTRNG
- SlimSRP as a DSP with reconfigurable core and 160 KB SRAM
- Sample Rate Converter (SRC) as a hardware accelerator
- Serial Flash Interface with execution-in-place (XiP) capability
- JTAG/SW interface for debug

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.2 Subsystems

4.2.1 Memory

Figure 3 illustrates the memory architecture of S1SBP6A. The system and its peripherals are little-endian format.



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Figure 3 Memory Architecture of S1SBP6A

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4.2.2 Micro Direct Memory Access Controller

S1SBP6A utilizes ARM PrimeCell Micro Direct Memory Access Controller (µDMAC) (PL230) as a Direct Memory Access Controller (DMAC). µDMAC is an AMBA compliant SoC peripheral that is developed, tested, and licensed by ARM. It is a very low gate count DMAC that is compatible with the AMBA AHB-Lite protocol.

µDMAC supports the following features:

- Compatible with AHB-Lite for the DMA transfers
- Compatible with APB for programming the registers
- Single AHB-Lite master for transferring data using a 32-bit address bus and 32-bit data bus
- 32 DMA channels
- Each DMA channel with dedicated handshake signals
- Each DMA channel with a programmable priority level
- Each priority level arbitrates using a fixed priority that is determined by the DMA channel number
- Multiple transfer types:
  - Memory-to-Memory
  - Memory-to-Peripheral
  - Peripheral-to-Memory
- Multiple DMA cycle types
- Multiple DMA transfer data widths
- Access to primary, alternate, and channel control data structure of each DMA channel
- Little-endian format for storing all channel control data in system memory
- DMA transfers using the SINGLE AHB-Lite burst type
- Equal destination data width to the source data width
- Programmable number of transfers in a single DMA cycle from 1 to 1024
- Greater transfer address increment than the data width
- Single output to indicate when an ERROR condition occurs on the AHB bus.

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4.2.3 Slim Samsung Reconfigurable Processor (SlimSRP)

S1SBP6A utilizes Slim Samsung Reconfigurable Processor (SlimSRP) as an energy-efficient Digital Signal Processor (DSP). The architectural simplicity of SlimSRP processor leads to very small implementation which allows wearable devices with very low power consumption.

SlimSRP supports the following features:

- 32-bit RISC architecture
- 3-issue VLIW processor
  - Slot 1: control, basic ALU, LOAD/STORE
  - Slot 2: ALU, MAC, LOAD/STORE
  - Slot 3: ALU, MAC
- Register Files
  - Register File (32-bit × 32)
  - Immediate Register File (28-bit × 4)
  - Special Register File
- Complex Instruction for DSP applications
- Single-cycle dual 32-bit × 16-bit multiply-and-add operation
- Bi-directional shift-and-saturate/round
- Bit mask, bit concatenation, leading zero detection
- Memory System
  - Tightly Coupled Memory
  - Data Scratch Pad Memory
- AHB Slave Interface

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4.2.4 Sample Rate Converter

Sample Rate Converter (SRC) is a sort of hardware accelerator which performs resampling data to align the sampling rates of data streams. It supports decimation and interpolation methods.

Following are the two operating methods of rate conversion:

- When resampling is done using (integer) decimation, then you can use a simple algorithm. In that case, the division of the input and output sampling rate is an integer number.
- When input and output frequency are not an integer multiple of each other, then you must perform the interpolation. The output samples are not reusing the input samples and needs to be calculated by means of interpolation.

4.2.5 General Purpose I/O

GPIO is a General Purpose I/O interface unit. It provides a 16 bits I/O interface for each GPIO port with the following properties:

- Programmable interrupt generation capability
- Bit masking support using address values
- Registers for alternate function switching with pin multiplexing
- Thread safe operation by providing separate set and clear addresses for control registers
- Inputs are sampled using a double flip-flop to avoid meta-stability issues.

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4.2.6 Timers

4.2.6.1 General Purpose Timer

S1SBP6A utilizes Timer of CMSDK (ARM DDI 0479B r0p0).

General Purpose timer is a 32 bits down-counter with the following features:

- The interrupt request signal is generated when the counter reaches 0. The interrupt request is held until it is cleared by writing to the INTCLEAR Register.
- If the timer count reaches 0 and the software clears a previous interrupt status at the same time, then the interrupt status is set to 1.
- Timer can be enabled when the 0 to 1 transition of external input signal occurs.
- The external clock must be slower than half of the peripheral clock, because it is sampled by a double flip-flop and then goes through edge-detection logic when the external inputs act as a clock.

4.2.6.2 PWM Timer

Basically, features of PWM Timer are the same as those of General Purpose Timer. Additionally, PWM Timer supports PWM function for motor driver.

PWM Timer supports the following features:

- Programmable period and high duration of PWM signal
- DC or AC motor PWM signal generation
- Programmable motor resonant frequency for AC motor

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4.2.6.3 Dual-Input Timer

S1SBP6A utilizes Dual-Input Timer of CMSDK (ARM DDI 0479B r0p0). Dual-input Timer consists of two programmable 32 bits or 16 bits down-counters that can generate interrupts when they reach 0. It can be used as 32 bits or a 16 bits counter with one of the following timer modes:

- Free-running
- Periodic
- One-shot

The operation of each timer module is identical. Dual-Input Timer provides access to two interrupt-generating, programmable 32 bits Free-Running Counters (FRCs). The FRCs operate from a common timer clock. Each FRC also has a prescaler that can divide down the common timer clock rate by 1, 16, or 256. This enables the count rate for each FRC.

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4.2.6.4 Watchdog

S1SBP6A utilizes Watchdog of CMSDK (ARM DDI 0479B r0p0). Watchdog is based on a 32 bits down-counter that is initialized from the Reload register. Watchdog generates a regular interrupt depending on a programmed value. The counter decrements by one on each positive clock edge of Watchdog clock.

Watchdog monitors the interrupt and asserts a reset signal when the counter reaches 0, and the counter is stopped. On the next Watchdog clock edge, the counter is reloaded from the Load register and the count-down sequence continues. If the interrupt is not cleared by the time that the counter next reaches 0, then Watchdog reasserts the reset signal. Watchdog applies a reset to a system during software failure to recover from software crashes.

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4.2.7 Universal Asynchronous Receiver Transmitter

Universal Asynchronous Receiver Transmitter (UART) supports three independent UART channels with asynchronous and serial input/output ports for general purpose.

UART supports the following features:

- Channel 0 to 2 with 64-byte FIFO
- All channels with interrupt-based operation
- All channels with DMA-based or Interrupt-based operation
- All channels with auto flow control with nRTS and nCTS
- Handshake transmit/receive

4.2.8 Inter-Integrated Circuit

Inter-Integrated Circuit (I2C) unit supports serial data communication with the following features:

- DMA-based or interrupt-based operation
- Separate transmission and reception of data
- Non-integer clock division
- Master and slave operation modes

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4.2.9 Serial Peripheral Interface

Serial Peripheral Interface (SPI) is the ARM PrimeCell PL022 Synchronous Serial Port (SSP). It enables synchronous serial communications with peripheral devices that have either a Motorola SPI compatible interface, Texas Instruments Synchronous Serial interface, or National Semiconductor Microwire interface. SPI can behave as a master or a slave interface and can perform parallel-to-serial conversion on data written to an internal 16-bit wide and 8-deep transmit FIFO. It also performs serial-to-parallel conversion on the serial input data and buffers it a receive FIFO which is 16-bit wide and 8-deep. SPI asserts interrupts to request transmit and receive buffers service to indicate an overrun or receive timeout condition in the receive FIFO.

4.2.10 Advanced Encryption Standard

S1SBP6A utilizes Advanced Encryption Standard (AES) to secure the personal biometric information and body signals. The AES supports Electronic Code Block (ECB) and Cipher Block Chaining (CBC) modes. It also operates with 128-bit or 256-bit key sizes.

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4.2.11 Digital True Random Number Generator

Digital True Random Number Generator (DTRNG) system generates random numbers from thermal noise of a chip. The generated random numbers are non-deterministic sequence of bits and unpredictable. It consists of Clock Divider, TRNG, Post Processor, and TRNG FIFO. The clock frequency of TRNG block should be less than 0.1 MHz for stable operation. You can adjust the TRNG clock frequency. The random bits from TRNG block can have a bias which means the random bits can have more ones than zeros or vice versa. So, DTRNG system includes Post Processor as a bias compensator. Post Processor supports four different bias compensation methods which are Bypass, Linear Feedback Shift Register (LFSR), Von Neumann (VN), and XOR. Random bit from Post Processor is gathered at TRNG FIFO sequentially. After FIFO is filled up, you can read 32-bit random data from FIFO at a time.

4.2.12 Real Time Clock

Real Time Clock (RTC) unit operates using the backup battery when the system power is off. When the power is off, the backup battery can store the time by Second, Minute, Hour, Date of the week, Day, Month, and Year data. The RTC unit works with an external clock source.

4.2.13 Flash Controller

Flash Controller consists of an internal 2 MB Flash and 2 KB Cache. Flash controller is connected with two AHB buses, System Bus for Flash IP and multiplexed Instruction/Data-Code bus for Flash Controller.

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Flash Controller supports the following features:

- 2X/4X prefetch mode for high-speed operation
- Error Correction Code (ECC) function for improvement in flash cell reliability
- Cache for improvement in NOR flash memory performance

4.2.14 Serial Flash Interface

Serial Flash Interface (SFI) enables communications with external serial flash with SPI protocol.

SFI supports the following features:

- Single/Quad mode
- Maximum 256-byte burst write
- AHB slave interface
- Serial flash interface commands

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4.2.15 Interrupts

S1SBP6A contains 119 interrupt request (IRQ) lines, one Non-Maskable Interrupt (NMI), and one event signal. *Table 14* describes the interrupt assignments. Cortex-M4 processors have an RXEV input signal. If software uses the Wait For Event (WFE) instruction to put the processor to sleep, then the received event at RXEV wakes up the processor. The RXEV is connected to DMA-done signal of the μDMAC. This enables the processor to wake up from WFE sleep when the DMA process finishes.

Table 14 Interrupt Assignments

IRQ/NMI	Device	IRQ/NMI	Device	IRQ/NMI	Device	IRQ/NMI	Device
NMI	Watchdog	29	RTC2	59	DMA CH23	89	GPIO12
0	BOR	30	FPU	60	DMA CH24	90	GPIO13
1	UART0	31	SRP0	61	DMA CH25	91	GPIO14
2	UART1	32	SRP1	62	DMA CH26	92	GPIO15
3	UART2	33	Delay Monitor	63	DMA CH27	93	GPIO16
4	SPI0	34	AES	64	DMA CH28	94	GPIO17
5	SPI1	35	External Wakeup	65	DMA CH29	95	GPIO18
6	SPI2	36	DMA CH0	66	DMA CH30	96	GPIO19
7	SPI3	37	DMA CH1	67	DMA CH31	97	GPIO1A
8	SPI4	38	DMA CH2	68	GPIO0 Combined	98	GPIO1B
9	I2C0	39	DMA CH3	69	GPIO1 Combined	99	GPIO1C
10	I2C1	40	DMA CH4	70	GPIO2 Combined	100	GPIO1D
11	I2C2	41	DMA CH5	71	GPIO00	101	GPIO1E
12	I2C3	42	DMA CH6	72	GPIO01	102	GPIO1F
13	I2C4	43	DMA CH7	73	GPIO02	103	GPIO20
14	Timer0	44	DMA CH8	74	GPIO03	104	GPIO21
15	Timer1	45	DMA CH9	75	GPIO04	105	GPIO22
16	Timer2	46	DMA CH10	76	GPIO05	106	GPIO23
17	Timer3	47	DMA CH11	77	GPIO06	107	GPIO24
18	Timer4	48	DMA CH12	78	GPIO07	108	GPIO25
19	Timer5	49	DMA CH13	79	GPIO08	109	GPIO26
20	Timer6	50	DMA CH14	80	GPIO09	110	GPIO27
21	PWM Timer0	51	DMA CH15	81	GPIO0A	111	GPIO28
22	Dual Timer	52	DMA CH16	82	GPIO0B	112	GPIO29

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IRQ/NMI	Device	IRQ/NMI	Device	IRQ/NMI	Device	IRQ/NMI	Device
23	AFE Controller	53	DMA CH17	83	GPIO0C	113	GPIO2A
24	SRC	54	DMA CH18	84	GPIO0D	114	GPIO2B
25	DMA Error	55	DMA CH19	85	GPIO0E	115	GPIO2C
26	Embedded Flash	56	DMA CH20	86	GPIO0F	116	GPIO2D
27	RTC0	57	DMA CH21	87	GPIO10	117	GPIO2E
28	RTC1	58	DMA CH22	88	GPIO11	118	GPIO2F

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4.2.16 Phase-Locked Loop

Phase-Locked Loop (PLL) in S1SBP6A is a clock generation IP which synthesizes wide range frequency output. It consists of Phase-Frequency Detector (PFD), charge pump, Voltage-Controlled Oscillator (VCO), 6-bit pre-divider, 7-bit main-divider, 2-bit scaler, and Automatic Frequency Control (AFC).

The output frequency (F_{FOUT}) is related to the input frequency (F_{FIN}) using the following equations.

$$F_{FVCO} = \frac{m \times F_{FIN}}{p}$$

$$F_{FOUT} = \frac{m \times F_{FIN}}{p \times 2^s}, 1 \le p \le 63, 16 \le m \le 127, 0 \le s \le 3$$

Where, p, m, and s are the divide values for pre-divider, main-divider, and scaler respectively.

4.2.17 High Speed Oscillator

High Speed Oscillator (HSOSC) in S1SBP6A is a clock generation IP which is an RC oscillator with output frequency of 24.576 MHz. It can be trimmed during fabrication and can generate precise clock frequency by calibration with external 32.768 kHz clock optionally. HSOSC consists of oscillator core, current reference, and Frequency-Locked Loop (FLL).

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4.2.18 Low Speed Oscillator

Low Speed Oscillator (HSOSC) in S1SBP6A is a clock generation IP which is an RC oscillator with output frequency of 4.096 MHz. It can be trimmed during fabrication and can generate precise clock frequency by calibration with external 32.768 kHz clock optionally. LSOSC consists of oscillator core, current reference, and Frequency-Locked Loop (FLL).

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4.3 System Control

4.3.1 Startup Sequence

Figure 4 illustrates the startup sequence of S1SBP6A.

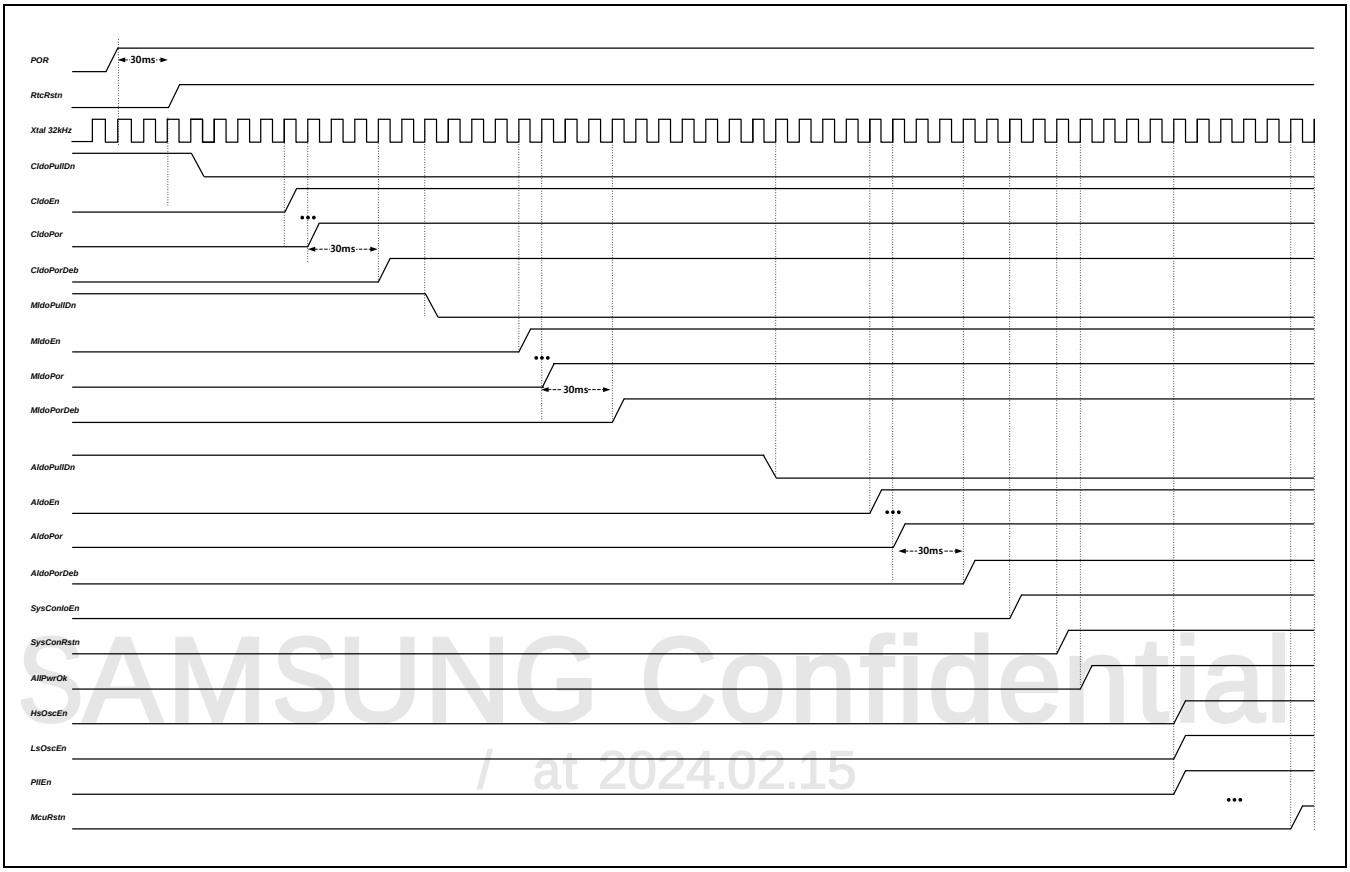


Figure 4 Startup Sequence of RTC Logic

The sequence begins when external 3 V powers, AVDD33, and DVDD33 are supplied. Power-on Reset (POR) used as the reset of RTC logic is released after AVDD33 is settled down, and it is de-bounced in 30 ms using 32.768 kHz clock from external crystal input. The de-bounced POR initiates power sequencer of RTC logic. Internal LDOs, CLDO, MLDO, PLDO, and ALDO are turned on sequentially. Each LDO outputs LDO POR indicating the readiness of LDO power supply. Those LDO PORs are also de-bounced in 30 ms for stable power supply operation. After releasing of the last POR from ALDO, IO Power Switches, which supply powers of normal GPIOs and isolated crystal IO, are turned on. Finally, reset toward System Controller is released. System Controller logic starts after releasing of the reset from RTC logic. The reset from RTC logic is de-bounced in 30 ms, and then internal oscillators, LSOSC and HSOSC, are turned on. MCU reset is released after ticking by number of several clocks.

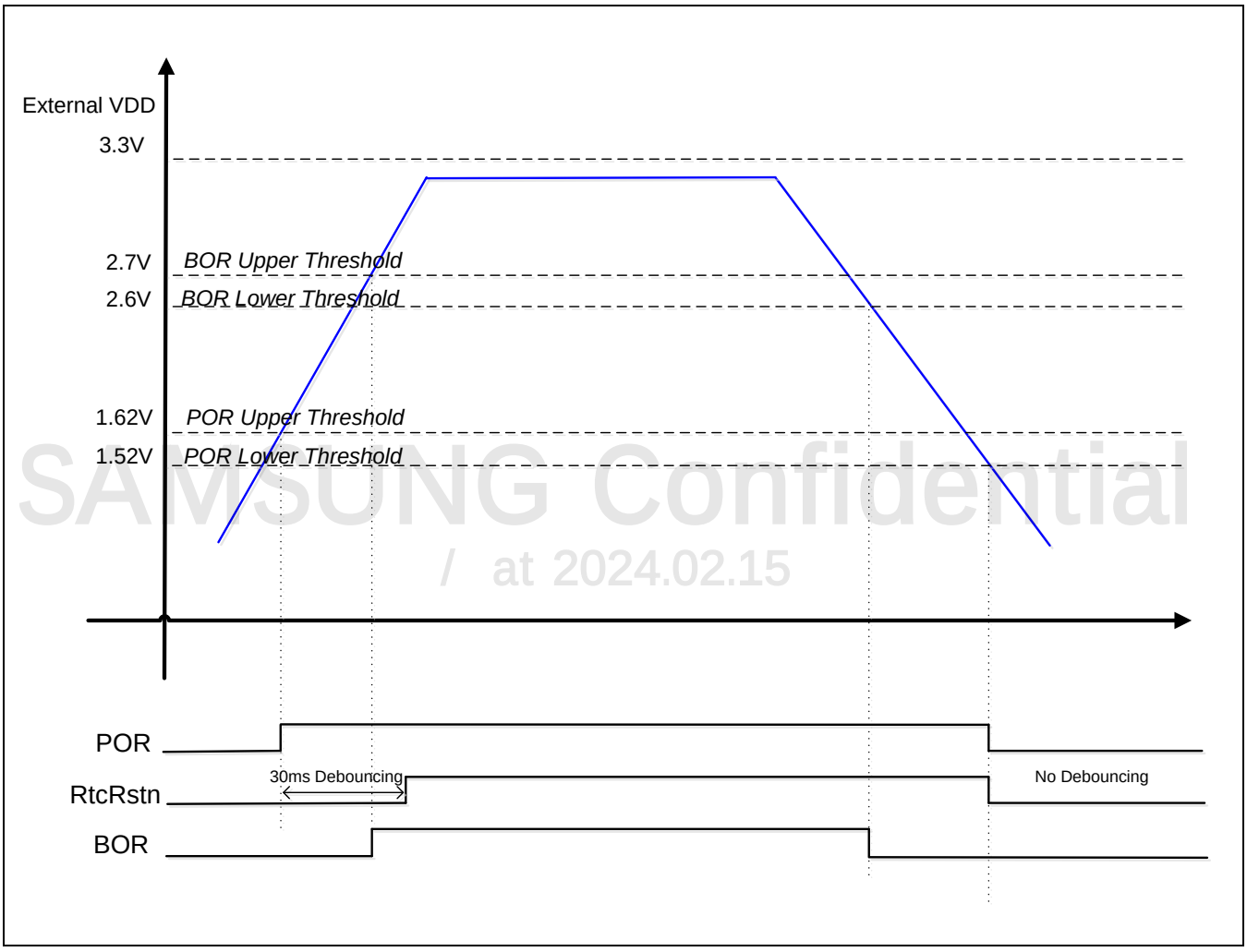
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4.3.2 Power-on Reset (POR) and Brown-out Reset (BOR)

BOR and POR have simple hysteresis architectures which have a high level reference and a low level reference. BOR detects a brown-out voltage when power supply is falling down and sends BOR state signal to RTC logic. In the same way, POR detects a power-on voltage when power supply is rising up and sends POR state signal to RTC logic.

As illustrated in *Figure 5* , BOR has upper threshold and lower threshold. If external power is 3 V with ± 10 % variation, the upper threshold of BOR is 2.7 V (2.6 V + 0.1 V) and the lower threshold is 2.6 V. The upper threshold of POR is 1.62 V, and in the same way, the lower threshold is 1.52 V (1.62 V -0.1 V).



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Figure 5 BOR and POR Threshold Levels and Reset Signals

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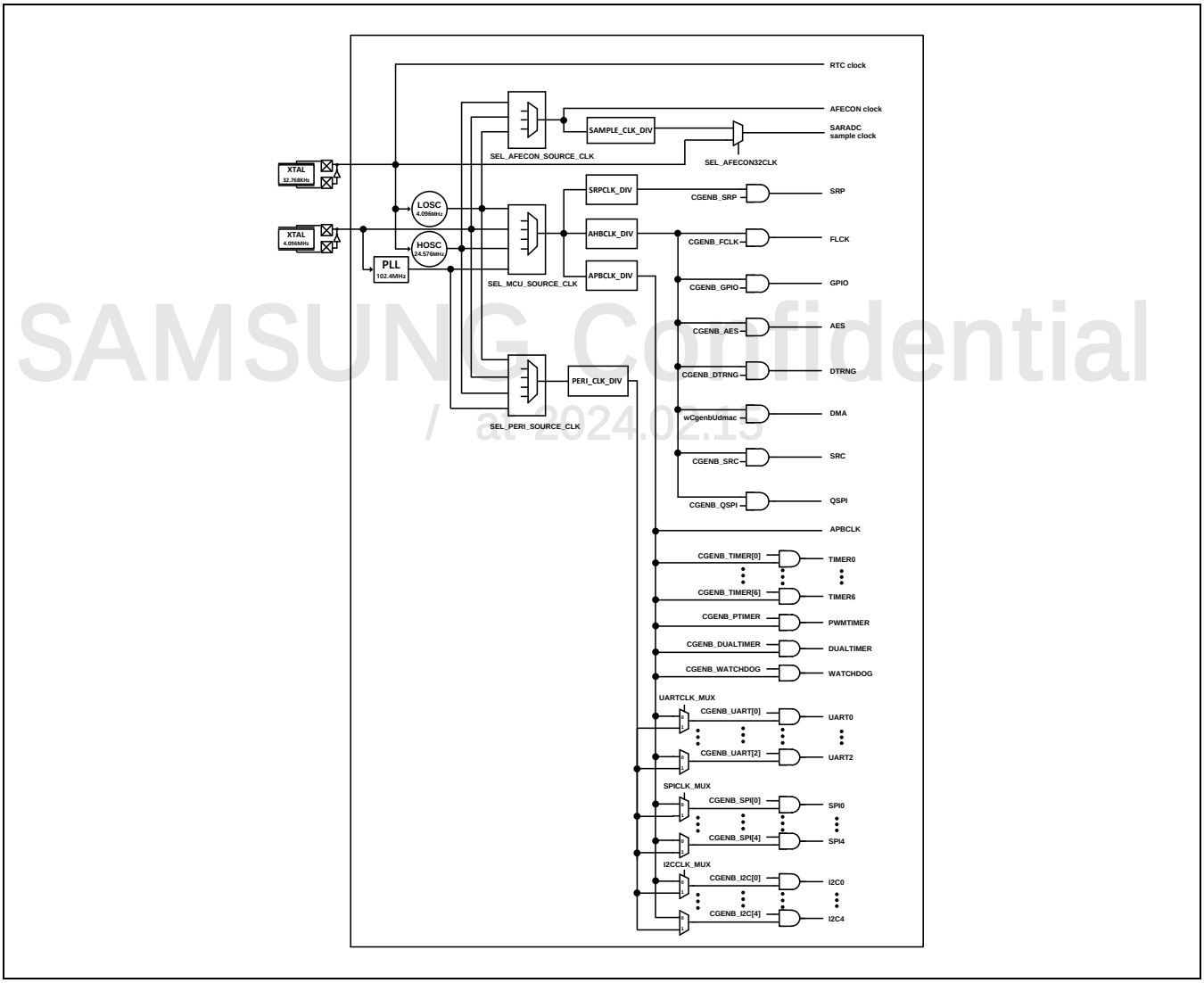
4.3.3 Clock Management Unit

S1SBP6A includes Clock Management Unit (CMU) which manages clock gating controls to reduce active current consumption.

CMU supports the following four features:

- Source clock selectivity for MCU, Peripherals, and AFE Controller
- Clock divider programmability of MCU, Peripheral, and SAR-ADC sampling clock
- Clock gating capability of separated clock domains
- Clock selectivity of serial interfaces

Figure 6 illustrates the clock tree of S1SBP6A.



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Figure 6 Clock Tree of S1SBP6A

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4.3.4 Power Management Unit

S1SBP6A includes Power Management Unit (PMU) which manages the modes of system operation. PMU executes the power on-off controls of sub-blocks and mode change control to reduce power consumption.

S1SBP6A has 22 power domains including one always-on domain and 21 power-gating or power-off domains. RTC block is always-on to perform real-time clocking function and managing startup sequence of the chip. The rest of the chip except RTC block is turned off in LPM0, the lowest power mode of S1SBP6A described in *Table 15* by shutting down the internal LDOs. MCU has 12 power domains including MCU system, Cortex-M4 core, Flash and Cache, and eight 32 KB SRAMs. Cortex-M4 core is retained under the low-power modes except LPM4. In these four low-power modes, all information is saved for the processor to continue the program which is stopped before power down. Power gating is not applied and only clocks are gated in LPM4. For the memories, Flash Cache and MCU SRAMs can be power-gated or retained. SRP includes six power domains including one SRP logic, one SRP TCM, and four SRP DMEMs. SRP logic supports only power-gating option and all of SRP SRAMs can be power-gated or retained. *Table 15* describes the modes of operation available in S1SBP6A.

Table 15 Low-Power Modes of S1SBP6A

Mode Name	Entry	Wakeup Sources	Descriptions
LPM0	LPM2 + PLDOEN_CTRL = 1 + PLDOPULLDN_CTRL = 0 + RTCMODEREADY = 1	BMODE RTC interrupts	RTC on, Others off
LPM1	LPM2 + STANDBYREADY = 1	GPIO00-1F pins Timer0 interrupt Uart0 interrupt SPI0 interrupt I2C0 interrupt	RTC/SYSCON on, Flash sleep, Others off
LPM2	SYSPWR_CTRL = 1 + PMU_CTRL = 1 + SLEEPDEEP =1 + WFI	Timer0 interrupt Uart0 interrupt SPI0 interrupt I2C0 interrupt DMA requests	MCU system retention, Cortex core retention, Flash on
LPM3	PMU_CTRL = 1 + SLEEPDEEP =1 + WFI	Timer0 interrupt Uart0 interrupt SPI0 interrupt I2C0 interrupt General Interrupts	MCU system on, Cortex core retention
LPM4	PMU_CTRL = 1 + WFI	Timer0 interrupt Uart0 interrupt SPI0 interrupt I2C0 interrupt General interrupts	Cortex core clock gating

NOTE: SLEEPDEEP is the [2] bit field of System Control Register (SCR) of Cortex-M4 core. For more information on SLEEPDEEP, refer to Cortex-M4 Devices Generic User Guide, *http://infocenter.arm.com/help/topic/com.arm.doc.dui0553a/DUI0553A_cortex_m4_dgug.pdf*

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Following are the modes of operation available in S1SBP6A:

- LPM0 (Low-Power Mode 0)

S1SBP6A supports ultra-low power mode named LPM0. Only RTC block is alive and all other blocks are turned off in this mode. The chip enters LPM0 by setting PLDOEN_CTRL, SYSPWR_CTRL, SLEEPDEEP, and RTCMODEREADY bits into 1 and PLDOPULLDN_CTRL bit into 0 and executing WFI or WFE system function. Processor wakes up by toggling of BMODE or RTC Interrupts.

- LPM1 (Low-Power Mode 1)

S1SBP6A enters into LPM1 by setting SYSPWR_CTRL, SLEEPDEEP, and STANDBYREADY bits into 1, and executing WFI or WFE system function. At the same time, GPIO00 to GPIO1F pins should be low. In LPM1, Only RTC block and system control logic are alive. Flash is in sleep mode in this mode. Processor wakes up when any changes of from GPIO00 to GPIO1F pins or any Wait-Peri interrupts occur.

- LPM2 (Low-Power Mode 2)

S1SBP6A enters into LPM2 by setting SYSPWR_CTRL and SLEEPDEEP bit into 1 and executing WFI or WFE system function. In LPM2, Cortex core and MCU system block goes to retention mode. In this mode, processor wakes up by Wait-Peri interrupts from UART0, I2C0, Timer0, SPI0, and DMA requests.

- LPM3 (Low-Power Mode 3)

S1SBP6A enters into LPM3 by setting SLEEPDEEP bit into 1 and executing WFI or WFE system function. In this mode, Cortex core goes to retention state where all the data of flip-flops are retained. In LPM3, processor wakes up by any interrupts described in *Table 14* or wakeup events.

- LPM4 (Low-Power Mode 4)

S1SBP6A enters into LPM4 by WFI or WFE system function. Core clock is gated in LPM4 and processor wakes up when any interrupts described in *Table 14* or wakeup events occur.

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5 Analog Front End

This chapter includes the following sections:

- Overview
- Functional description

5.1 Overview

Figure 7 illustrates the block diagram of Analog Front End (AFE).

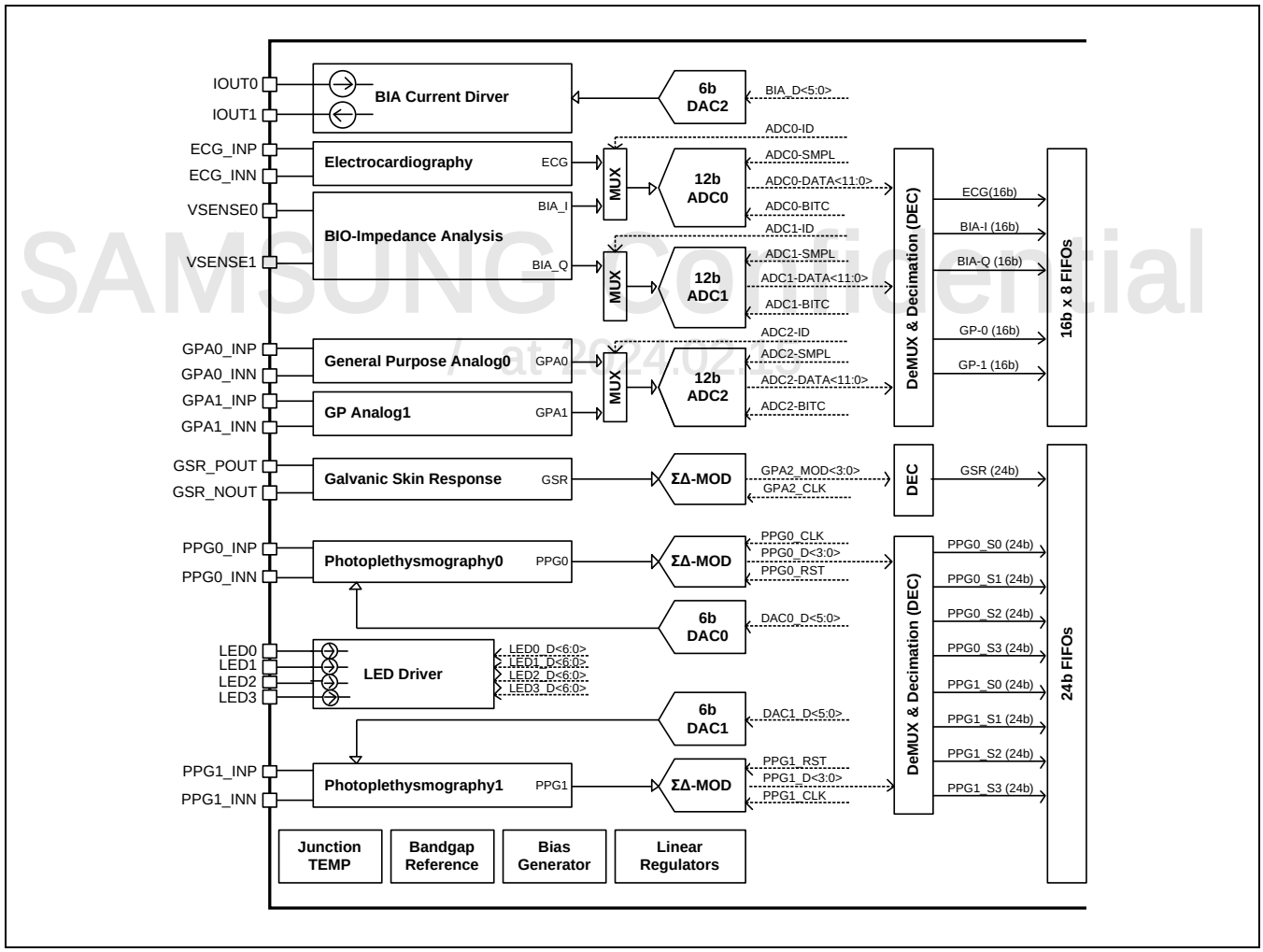


Figure 7 Block Diagram of Analog Front End

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S1SBP6A integrates multiple low-power AFEs to support wide range of bio-sensor applications. It includes an ECG readout, a BIA readout, a GSR readout, and two PPG readouts along with four LED drivers. In addition, several analog interfaces are provided for general usage. These includes a time multiplexed 12 bit ADC with supporting voltage and current inputs. *Figure 7* illustrates the simplified top-level architecture of AFEs in S1SBP6A. The complete AFEs receive power either from a built-in regulator (ALDO) or from an external regulator (through the pins of AVDD33, AVDD33A, and AVDD33_LED). Necessary bias signals for AFEs are generated from a fine-tuned bandgap reference. Output data streams from different AFEs are synchronized with each other and it allows implementing accurate and reliable parameter calculations in digital domain.

Following are the main functional blocks and their description:

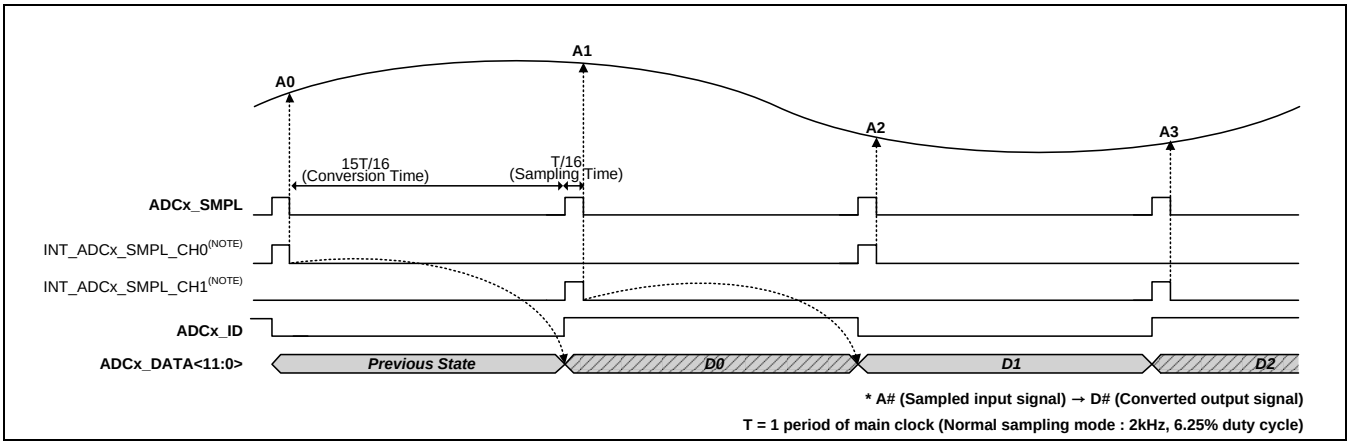
- The ECG readout measures bio-potential signals originated from heart. It supports high impedance dry electrodes. Moreover, a built-in HPF at inputs suppresses potential DC offset from electrodes in practice. The complete AFE is optimized for low-noise and low-power consumption. A built-in driver provides bias to the body through electrode.
- The AFE for PPG supports Photo Diode (PD) readout. Along with LED driver, the PD and the AFE measure the reflected or transmitted lights modulated by heartbeat. The AFE operates in a synchronous manner with the LED pulse. It separates up to four different signals per sample before digitization. These four signals from single PD are responses of different wavelengths of LEDs and ambient light. With the help of digital signal processing, real-time ambient light cancellation for PPG is possible which is essential for the applications of heart rate monitoring and SpO2 evaluation. Both PPG readout and LED driver provide wide range of programmability such as LED driving currents, pulse rate and pulse duty, readout gain, and input DC photocurrent cancellation capability. These flexibilities offer wide options for user to trade-off between power consumption and signal quality.
- The AFE for BIA predicts body impedance from measurement of voltage level in different body locations. An (externally) AC-coupled current source with programmable current-levels and frequency is utilized to inject current signals into the body through a pair of current electrodes. The fat inside the body impedes the injected AC current flow. By using another pair of (voltage) electrodes, the BIA AFE readouts voltage signal built across bio-impedance. For body composition analysis, the frequency of current signal can be adjusted from 1 kHz to 1 MHz with fine-grade resolution. The voltage signal received by BIA AFE can be demodulated either in rectified mode (envelope detection) or in I/Q mode (quadrature detection) according to the user preference.
- The AFE for GSR is implemented to measure skin conductance of human body. The electrical characteristics of the skin have a strong correlation to the sympathetic nervous system and therefore, GSR is an important indication of psychological or physiological arousal. The GSR AFE supports two electrodes placed on the skin. These electrodes are driven by either DC or AC voltage signal with variable scale generated by the GSR AFE. The derived current corresponding to the skin conductance is then converted into voltage signal by internal resistors. A 24-bit ADC is used for digitization.
- S1SBP6A includes several general purpose (GP) analog interfaces to additionally support wide range sensor applications such as measuring temperature, pressure, weight, or light intensity. The GP analog interfaces in S1BP6A contain a 12-bit, up to 8 kSPS ADC cooperating with a highly programmable AFE (GP Analog0) that supports either voltage or current inputs, and a direct external inputs (integrated anti-alias filter, GP Analog1 multiplexing with GP Analog0).
- S1SBP6A also integrates digital control logic to cooperate with AFEs. The function of AFE control logic includes AFE channel separation from ADC outputs, decimation, and data buffering with either 16-bit or 24-bit data format with 16 depths.

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- S1SBP6A integrates three identical 12-bit ADCs. Depending on the states of ADCx_ID, each ADC converts two analog signals from AFEs in time multiplexed manner. ADC sample rate is determined by clock signal of ADCx_SMPL and the rate can be 2 kHz as illustrated in *Figure 8* for normal sampling mode with a duty cycle = 6.25 % or 16 kHz as illustrated in *Each ADC uses* the internal sampling clocks.

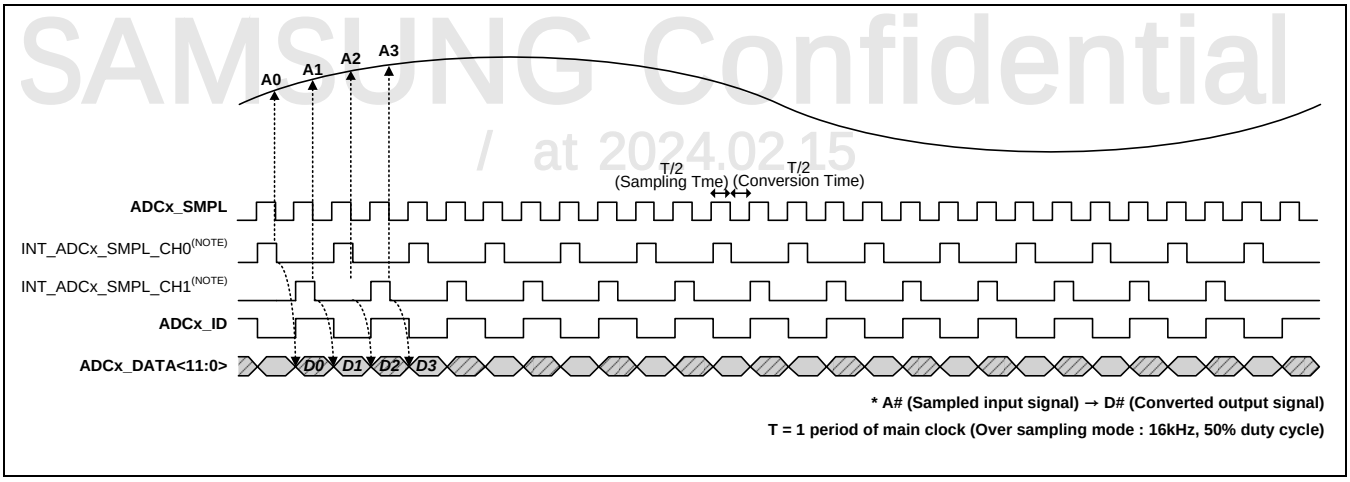
- Figure 9 for oversampling mode with a duty cycle = 50 %.



NOTE: Each ADC uses the internal sampling clocks.

Figure 8 Timing Diagram of 12-bit ADC Operating in Normal Sampling Mode

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NOTE: Each ADC uses the internal sampling clocks.

Figure 9 Timing Diagram of 12-bit ADC Operating in Oversampling Mode

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5.2 Functional Description

5.2.1 ECG Readout

5.2.1.1 Overview

ECG readout channel is a low-power integrated AFE dedicated for bio-potential signal acquisition. The supply voltage of ECG readout is 1.1 V and it is regulated by the built-in ALDO. The ECG readout channel illustrated in *Figure 10* consists of the following components:

- Input switch matrix that supports the following functions
  - ECG signal acquisition with bipolar electrode montage
  - DC offset measurement of ECG readout chain
  - Built-in square wave signal generation for the readout channel gain measurement.
- A built-in input High-Pass Filter (HPF) to avoid electrode DC offset amplification. This HPF can be bypassed to allow external HPF application. An integrated pair of 20 MΩ resistors provides an option for external HPF with only series capacitors.
- A fully differential chopper-modulated Instrumentation Amplifier (IA) with max. 10 mV_{p-p} input signal swing.
- A Programmable Gain Amplifier (PGA) stage to provide additional signal amplification of from 0 dB to 15.6 dB. This stage also embeds a passive second-order Low Pass Filter (LPF) with –3 dB bandwidth from 75 –250 Hz, to suppress chopping spike signals from IA stage as well as to provide first-order anti-alias filter function for noise suppression.
- A channel buffer with optional (needs two external capacitors, each of 0.1 μF) HPF with –3 dB bandwidth of 0.2 Hz. The register value of ECG_HPF2_EN enables this feature.
- A 12 bit ADC for ECG signal digitization. This ADC can be operated at a rate of either 1 kSPS or 8 kSPS along with digital decimation to provide improved Effective Number of Bits (ENOB).
- A built-in 3 bit DAC with output driver to bias human body through electrode. The output voltage can be varied from 330 –715 mV. The maximum output current drivability of body bias is 10 μA.
- *Figure 10* illustrates the block diagram of ECG readout channel.

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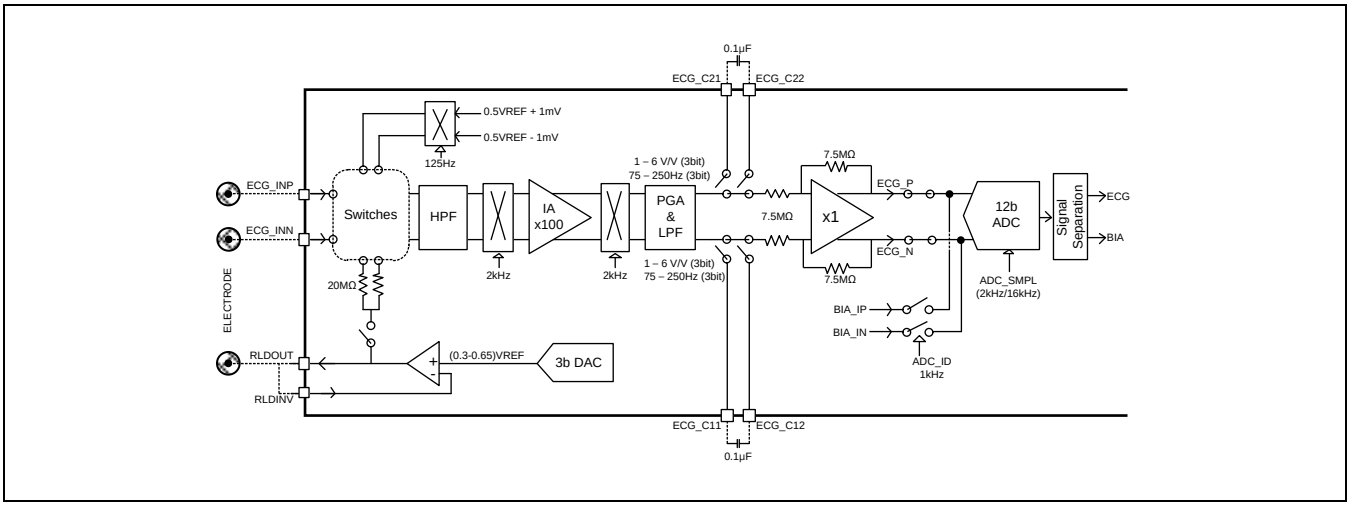


Figure 10 Block Diagram of ECG Readout Channel

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5.2.1.2 ADC Timing Diagram

ADC0 samples ECG and BIA output signals and converts them into 12 bit data stream (ADC_DATA in *Figure 11*). Sampling frequency of ADC can be configured to either 2 kHz or 16 kHz. To differentiate ECG and BIA signals, ADC_ID0 separates the sampled data at the output. When the ADC outputs the signal at a rate of 16 kSPS, a digital decimation is applied to the ADCx_DATA to form 16 bit data format. *Figure 11* illustrates the ADC timing diagram for ECG readout channel.

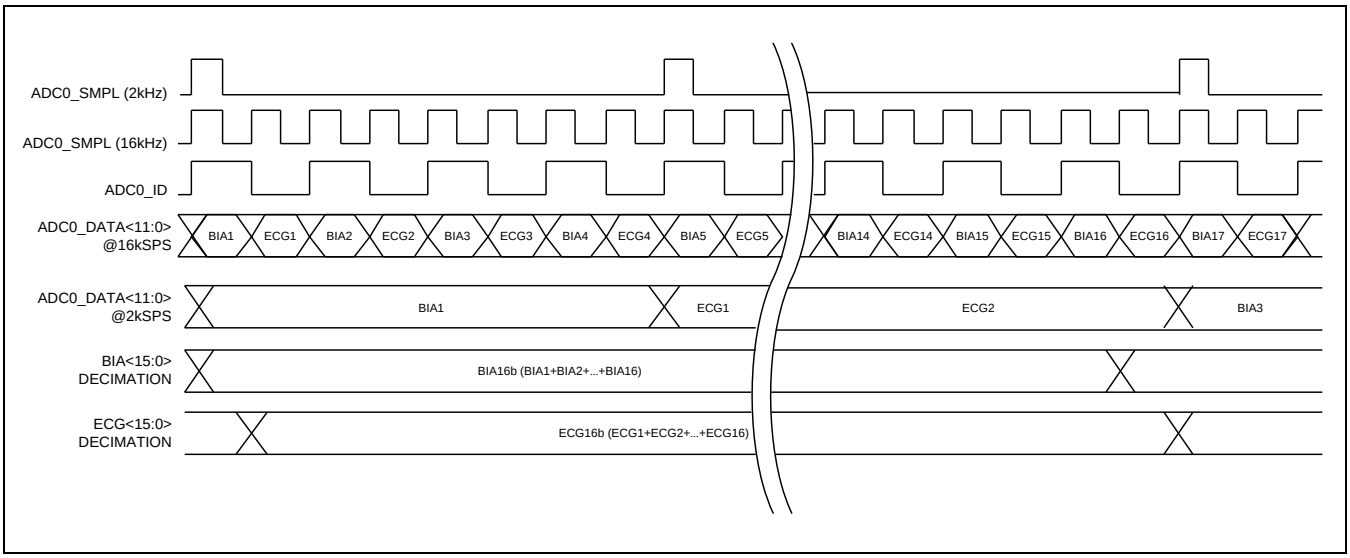


Figure 11 ADC Timing Diagram for ECG Readout Channel

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5.2.2 PPG Readout

5.2.2.1 Overview of PPG Measurement

PPG system irradiates light of a particular wavelength to the skin and senses reflected light which is modulated by the change in blood volume flowing through the capillaries and veins. By doing so, obtaining meaningful health parameters such as heart rate and SpO2 is possible. Following is the basic measurement procedure of PPG:

- Irradiate LEDs synchronized to the AFE sampling sequence.
- Use Photo Diode sensor to detect the reflected light and convert the light signals into photo current.
- Use PPG AFE to convert the photo current signal into voltage and then use an Incremental ADC (IDC) to digitize it.
- Apply the necessary digital signal processing to restore DC level of PPG readout and then calculate the interested health parameter.

The PPG system consists of four transmitters (LED drivers) and two channel of receivers (PPG0, PPG1) as illustrated in *Figure 12*. The supply voltage of LED driver is 3 V supplied from the pin of AVDD33_LED. The supply voltage of PPG AFE is 3 V supplied from the pin of AVDD33A, and the supply voltage of PPG IDC is 2.5 V which is regulated by a built-in regulator. LED driver needs to drive LEDs for the shortest time possible to reduce overall power consumption. S1SBP6A allows LED light controlling time to vary from 50 μs –2.5 ms with 13-bit register control. Each PPG readout channel is composed of a TIA, a sample-and-hold (SNH) stage, and an IDC. The TIA receives photo current from PD. To enlarge dynamic range input photo current in the presence of large ambient light interference, a differential 6 bit current DAC is integrated at the inputs of TIA in each readout channel to cancel out unwanted portions of photo current and restore the DC output level of TIA. Each channel of PPG readout supports 14 noise-free bit conversion when DCC is not in operation. Timing of LED driver is synchronized to that of PPG0, and PPG1 can be used additionally when two PDs are needed.

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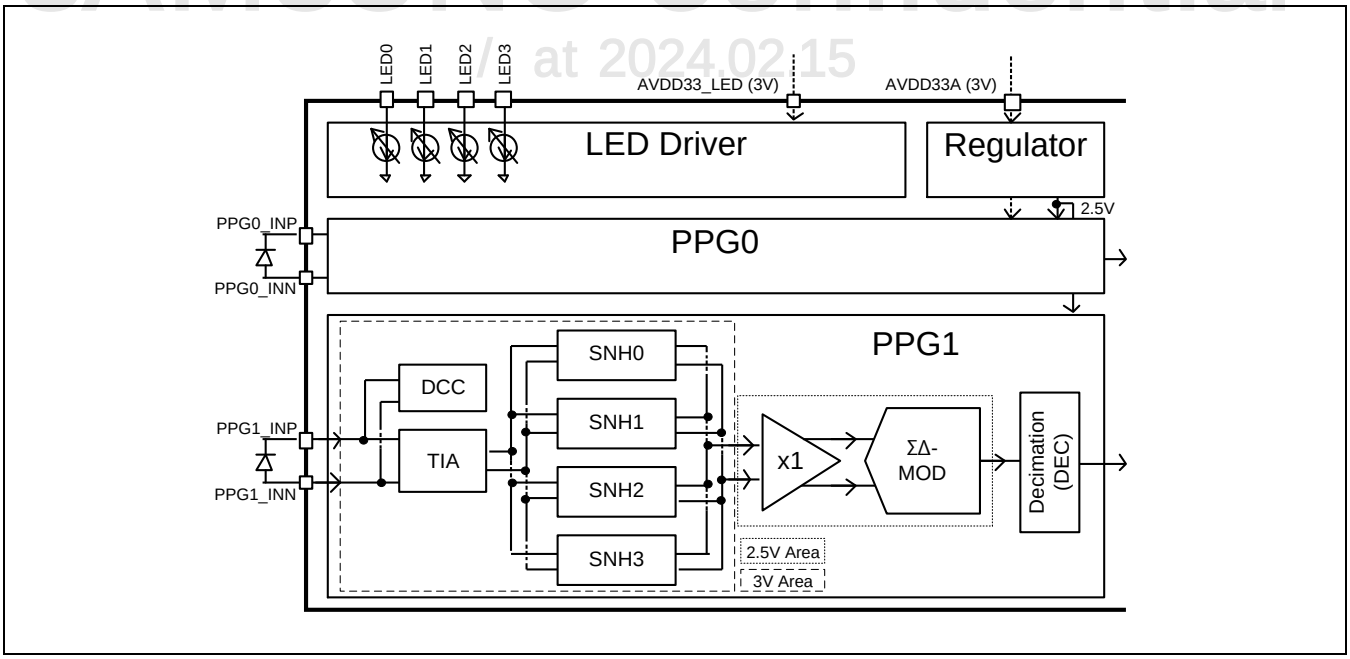


Figure 12 Block Diagram for PPG Readout Channel

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5.2.2.1.1 PPG System Timing Diagram

The PPG readout front-end has four parallel sample and hold (SNH) paths for each readout channel. As illustrated in *Figure 13*, four time slots can be allocated within one pulse repetition period (PPG_PRT). These four time slots can be assigned to the time period of turning on LEDs or the time period for sampling ambient light. For example, you can use two LEDs only within a single period of PPG_PRT. In this case, the slots for LED1 and slots for LED3 can be utilized to sample ambient signals such as AMBIENT1 and AMBIENT2, respectively. You can also use three LEDs within a single period of PPG_PRT. Similarly, AMBIENT3 replaces the slots for LED3 accordingly. Note that the rising edge of PPG sampling clock signal (PPG_SAM[3:0]) delays corresponding LED pulse (LED0, LED1, LED2, or LED3). This delay period encounters LED turn on time and the time required for TIA output sufficiently settling down. In a consequence, correct signal sampling is preceded during High state of PPG_SAM[3:0]. Data conversion process by the IDC stage is then initiated after the PPG_IDCRST goes "Low" and PPG_CONVT goes "High". Finally, 24-bit data is streamed out after PPG_CONVT becomes "Low". As illustrated in *Figure 13*, when there is no power-down period, after the value of register PPG_EN becomes "High", four TIMESLOTs (TIMESLOT0 - TIMESLOT3) are sequentially repeated within a pulse repetition cycle. As illustrated in *Figure 14*, the PPG can be powered-down by the state of PPG_IDC_PDN and PPG_AFE_PDN for low-power operation. PPG_PRT and PPG_CONVT determines the time of power-down period. IDLE, TIMESLOTS, TIMELSLOT1-3, TIMESLOTE, and POWERDOWN phase are sequentially repeated within a pulse repetition cycle. The register values of PPG_IDC_PDB_EN and PPG_AFE_PDN_EN enable the power-down option.

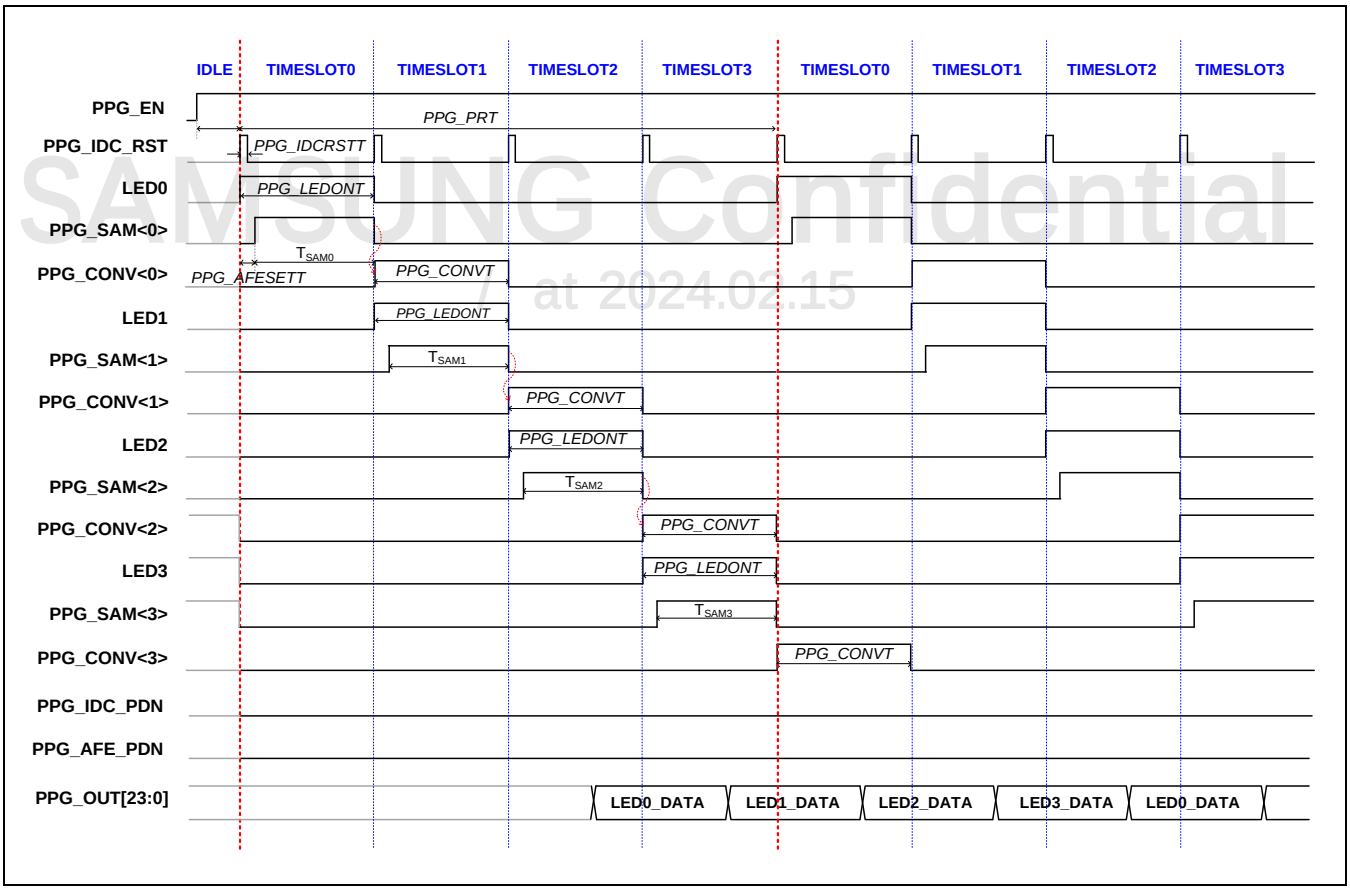
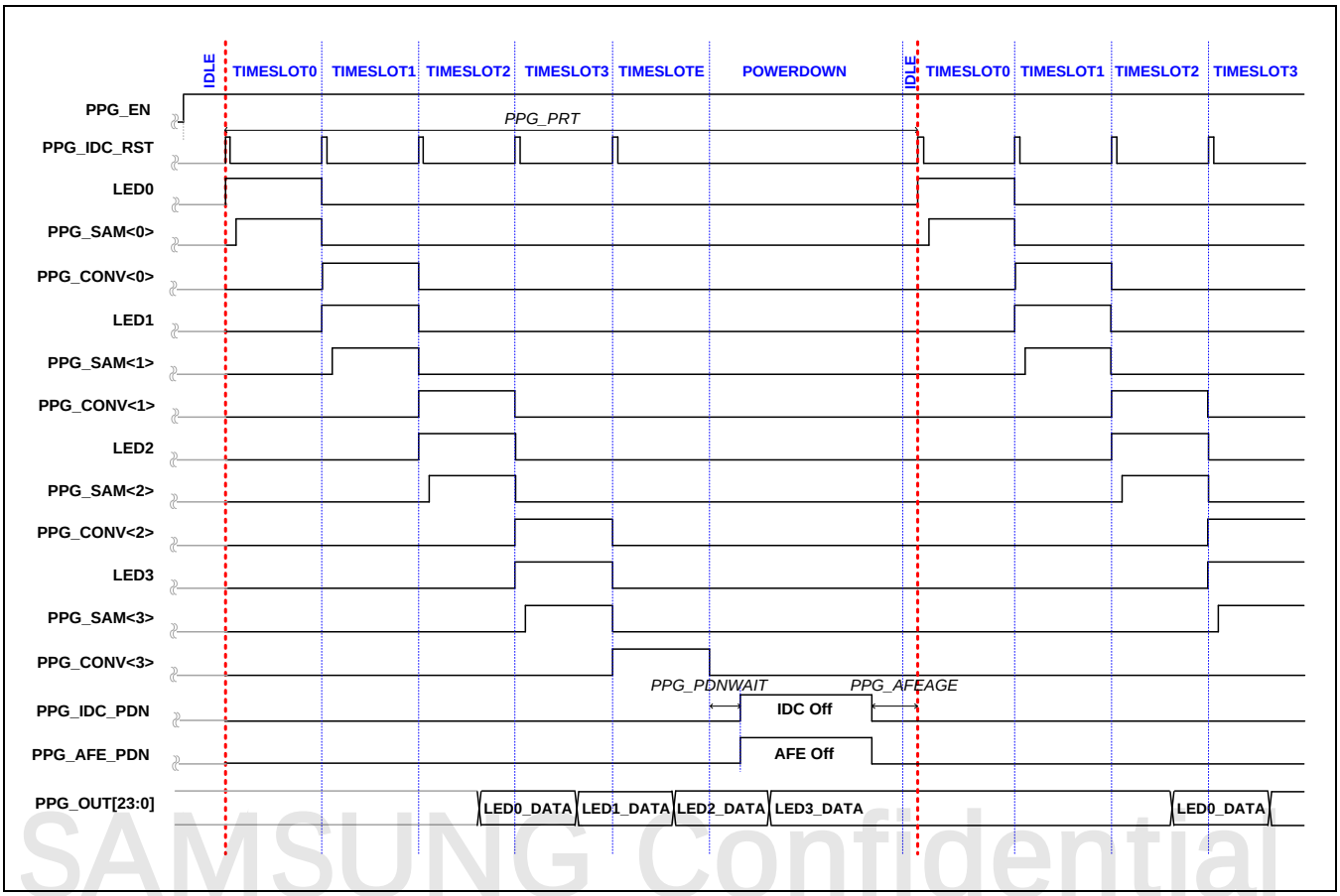


Figure 13 Timing Diagram for PPG Readout Channel

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Figure 14 Timing Diagram for PPG Readout Channel with Power-Down Period

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If PPG_PRT is 5 times longer than PPG_CONVT, PPG readout front end can be completely turned off to save power consumption. Use the following formula to determine the time to turn off the PPG readout (AFE Off, IDC Off).

$$\mathbf{T_{POWERDOWN}\ (sec) = PPG_PRT - 5 \times PPG_CONVT - PPG_PDNWAIT - PPG_AFEAGE}$$

The PPG readout front-end controls PRT, IDC data rate, LED on time, IDC reset time, and power down time. Register settings for timing control are described in *Table 16*.

Table 16 Timing Controls for PPG

Register	Bit	Default (HEX)	Description
PPG_PRT	17:0	0x1000	PPG Pulse Repetition Time (PRT) control. • Pulse repetition time (sec) = PPG_PRT/2.048 MHz • Pulse repetition frequency (Hz) = 2.048 MHz/PPG_PRT Minimum allowable PRT is limited to 1 ms (PPG_PRT = 0x800). Maximum allowable PRT is limited to 10 ms (PPG_PRT = 0x5000).
PPG_CONVT	12:0	0x200	PPG data conversion time control.

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Register	Bit	Default (HEX)	Description
			• Data conversion time (sec) = PPG_CONVT/2.048 MHz Maximum allowable data conversion time is limited to 25 % of PRT.
PPG_LEDONT	12:0	0x00CC	PPG on time control. • LED on time (sec) = PPG_LEDONT/2.048 MHz • PPG_LDEONT ≤ PPG_CONVT Maximum allowable LED on time is limited to 2.5 ms (PPG_LEDONT = 0x1400).
PPG_IDCRSTT	5:0	0x21	PPG IDC reset time control. • IDC reset time (sec) = PPG_IDCRSTT/2.048 MHz • PPG_IDCRSTT ≤ PPG_CONVT
PPG_PDNWAIT	8:0	0x29	PPG IDC and AFE power down wait time control. • Power down wait time (sec) = PPG_PDNWAIT/2.048 MHz
PPG_AFEAGE	8:0	0x00	PPG IDC and AFE aging time control. • AFE aging time (sec) = PPG_AFEAGE/2.048 MHz

5.2.2.2 LED Driver

LED driver is an integrated TX dedicated for PPG readout. The block diagram of LED driver illustrated in *Figure 15* consists of the following components:

- Bias block for current DACs. This block generates bias current for current DAC which can change the bias current by controlling the register value of LED_SEL_BIAS.
- Four 7-bit current DACs for two green LEDs, IR LED, and red LED (Current DAC0 and DAC1 are dedicated for driving green LEDs and Current DAC2 and DAC3 are dedicated for driving Red LED or IR LED). Each current DAC supports the options to adjust amount of output current to accommodate to various light intensity requirements of LEDs. Note that output currents of DAC0 ~ DAC3 are separately controlled.
- Digital logic to control LED driver. This logic converts digital signal from controller (LED_CTRLx[6:0] where x denotes 0, 1, 2, or 3, and LED_I_ENLED[3:0]).

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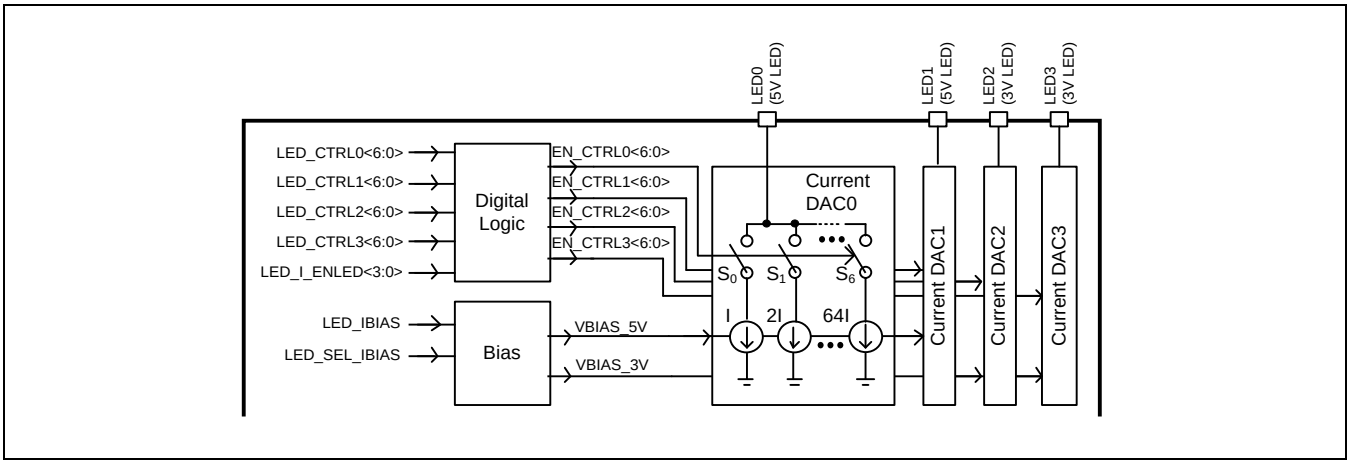


Figure 15 Block Diagram of LED Driver

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LED driving current is set by the register value of LED_CTRLx, where x can be 0, 1, 2, or 3. Use the following equation to calculate $IDAC_N$.

$$\mathrm{IDAC}_N(\mathrm{a}) = \frac{0.05 \times (2 - \mathrm{LED_SEL_BIAS})}{128} \times \mathrm{CTRL}_N$$

Where, $CTRL_N$ is the register value of LED_CTRLx. The minimum value of $IDAC_N$ is determined by the register value of LED_SEL_BIAS as described in *Table 17*.

Table 17 Minimum Value of Current DAC

LED_SEL_BIAS	Current DAC (mA/step)
0	0.78
1	0.39

Table 16 determines the register value of PPG_LEDONT that determines LED on time.

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5.2.2.3 PPG Readout Front-End

Two channels of PPG readout are integrated in S1SBP6A. The PPG readout signal chain illustrated in *Figure 16* consists of the following components:

- Input switch matrix support the following functions:
  - PPG input short function to measure TIA offset
  - DCC current test
  - TIA reset function to initialize TIA feedback capacitors
- TIA provides variable gain (trans-impedance) and bandwidth. The range of Gain is 10 kΩ to 2 MΩ with 3-bit register control. TIA supports a maximum of 100 pF of Photo Diode junction capacitance.
- Four SNHs integrated in each PPG channel allow separating signals (such as RED, RED_AMBIENT, IR, IR_AMBIENT) in analog domain. SNH provides low-pass filter function which has the bandwidth range of 40 - 500 Hz depending on the IDC data rate.
- DCC block for current sinking/sourcing at inputs of TIA. Maximum output current of DCC is 16 μA with 6-bit register control in 250 nA/step. This feature enlarges the input dynamic range. Notice that DCC current DAC at TIA inputs introduces additional noise to the complete PPG readout. For example, current noise of 100 pA_{rms} RTI (at 0.5 - 20 Hz, TIA gain is 500 kΩ) is evaluated by sourcing/sinking DCC with 8 μA. However, if the DCC current level is increased to 16 μA, the noise is doubled to 200 pA_{rms}.
- Shield driver is integrated to provide shielding external PPG inputs such as PCB input patterns. This feature minimizes the potential capacitive leakage happening at the pins of PPG_INP and PPG_INN.

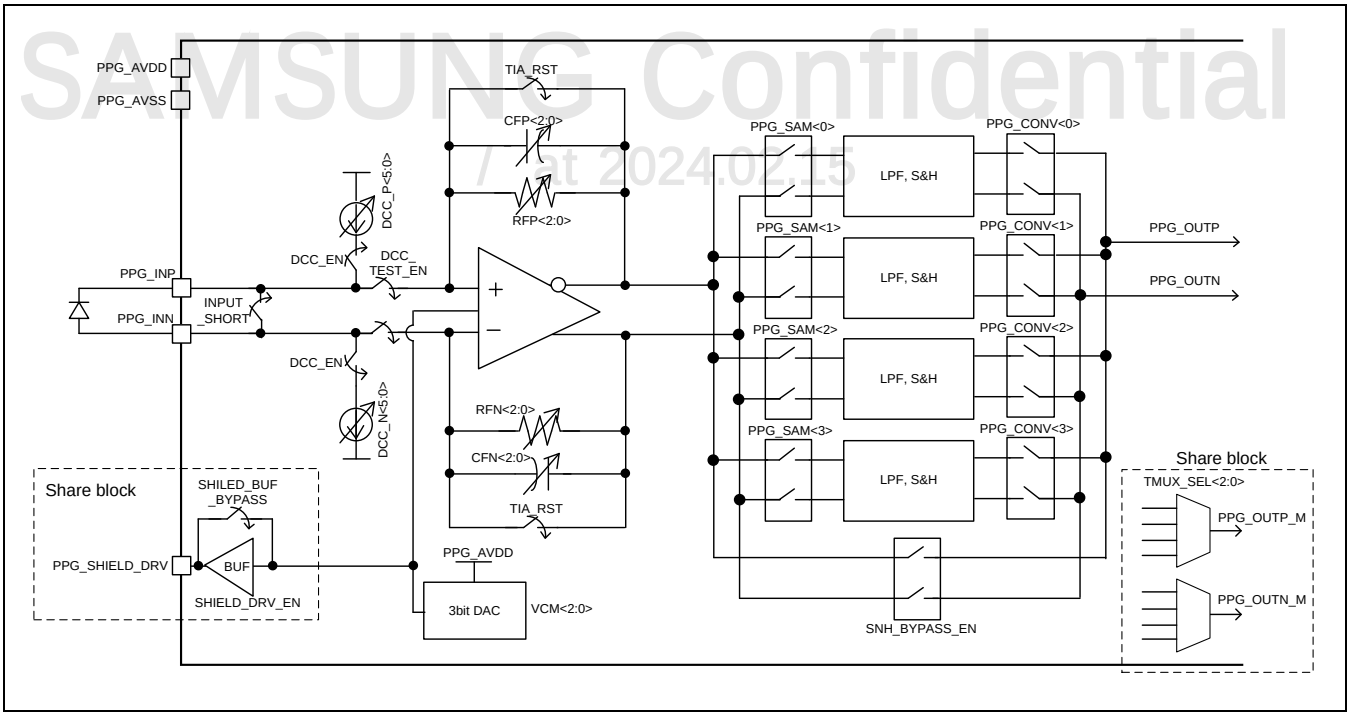


Figure 16 Block Diagram of PPG Readout Front-End

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5.2.2.3.1 TIA Gain Settings

Programmed value of RFP[2:0] and RFN[2:0] (the feedback resistor of the TIA) sets the value of TIA gain. The register value of PPG_RFP[2:0] and PPG_RFN[2:0] controls the RFP and RFN settings.

5.2.2.3.2 TIA Bandwidth Settings

TIA bandwidth settings are similar to TIA gain settings. The TIA bandwidth is set by programming the value of CFP[2:0] and CFN[2:0] (the feedback capacitance of the TIA). The product of RF and CF gives the time constant of the TIA and must be set approximately 20 % (or less) of the LED or sampling pulse durations. This choice of time constant allows the TIA to effectively pass the incoming pulses from the PD. The register value of PPG_CFP[2:0] and PPG_CFN[2:0] controls CF.

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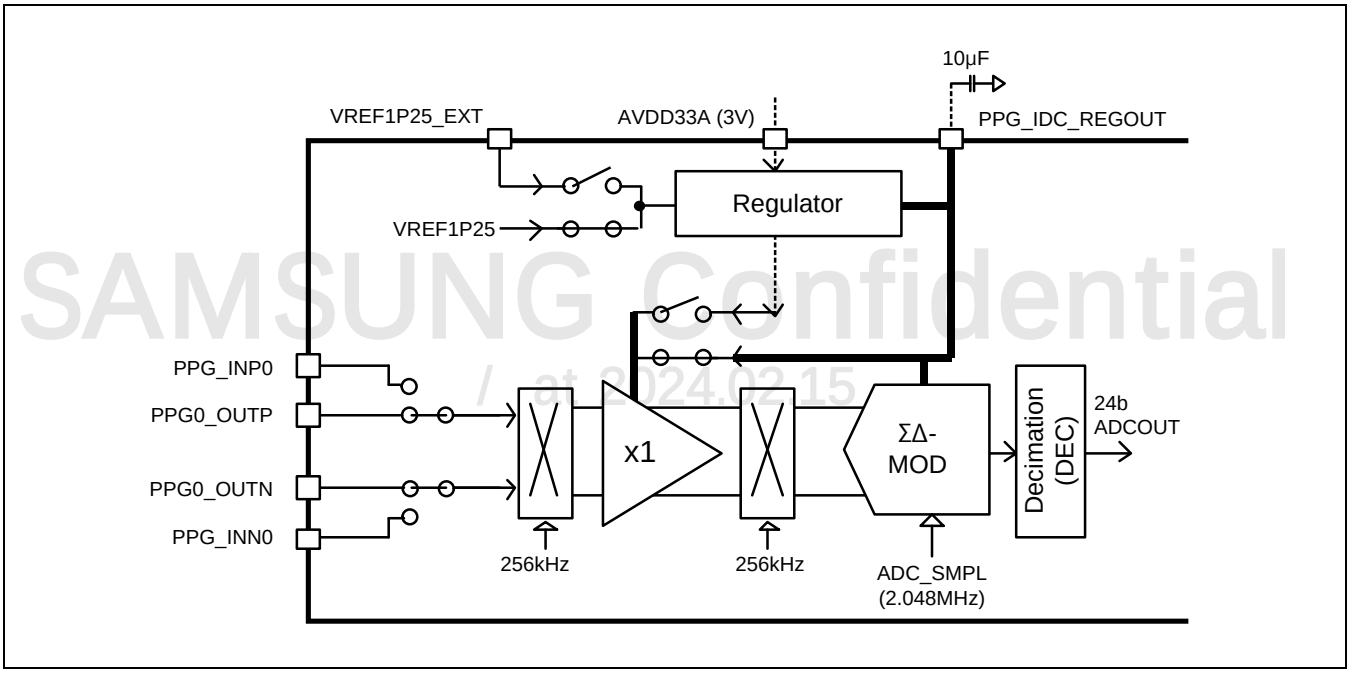
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5.2.2.4 Incremental ADC with Digital Decimation

5.2.2.4.1 Overview

IDC is integrated to support PPG AFE. The IDC incorporates voltage buffers at inputs helping to interface with high impedance sensors. The IDC operates with 3 V supply voltage (from the pin of AVDD33A) with internal voltage regulation, and outputs 24-bit data stream at a rate of 4 kSPS. The block diagram of IDC signal chain illustrated in *Figure 17* consists of the following components:

- Built-in voltage regulator generates on-chip supply voltage of 2.5 V from an external 3 V voltage source. For the regulator, 10 μF external capacitor is recommended to filter out excessive noise. Optionally, the regulator can be turned-off and IDC can be directly supplied from an external voltage source through regulator output.
- Voltage buffer allows high impedance interface. It supports input signal up to 3.4 V_{p-p} with 2.5 V supply voltage and 3.8 V_{p-p} with 3.0 V supply voltage.
- IDC oversamples analog input signal at a rate of 2.048 MHz. A digital filter is embedded to provide variable IDC data rate of 250, 500, 1k, 2k, 4k (default), 8k, and 16 kSPS.



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Figure 17 Block Diagram of Incremental ADC

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5.2.2.4.2 Incremental ADC Control

As illustrated in *Figure 18*, IDC samples the input signals and converts them into 24 bit data stream (ADCOUT[23:0]). Sampling frequency of IDC (CLK) is 2.048 MHz and the register value of PPG_CONVT[12:0] adjusts the data rate from 250 SPS to 16 kSPS as described in *Table 18.*

Table 18 IDC Data Rate Control

PPG_CONVT<12:0>	IDC Data Rate (kSPS)
0000010000000	16
0000100000000	8
0001000000000 (Default)	4 (Default)
0010000000000	2
0100000000000	1
1000000000000	0.5
1111111111111	0.25

Before turning on IDC_EN, ALDO and reference circuits must be turned on and a delay of 4 ms is required for settling down the node of PPG_IDC_REGOUT as illustrated in *Figure 17* . IDC should be reset at every conversion. The reset time (Low state of RSTB) is 32 × CLK period (1/2.048 MHz) of ΣΔ MOD.

The IDC has 14-bit current control options such as ICTRL_INT1[3:0], ICTRL_INT2[3:0], ICTRL_INT3[3:0], and ICTRL_QUANT[1:0] that control the current consumption of analog blocks of IDC. The wake up time of IDC (T_{WAKE}) is 50 × CLK period. This time period (PPG_AFEAGE as illustrated in *Figure 14*) is required to control PPG_IDC_PDN signal. *Figure 18* illustrates the timing diagram of IDC.

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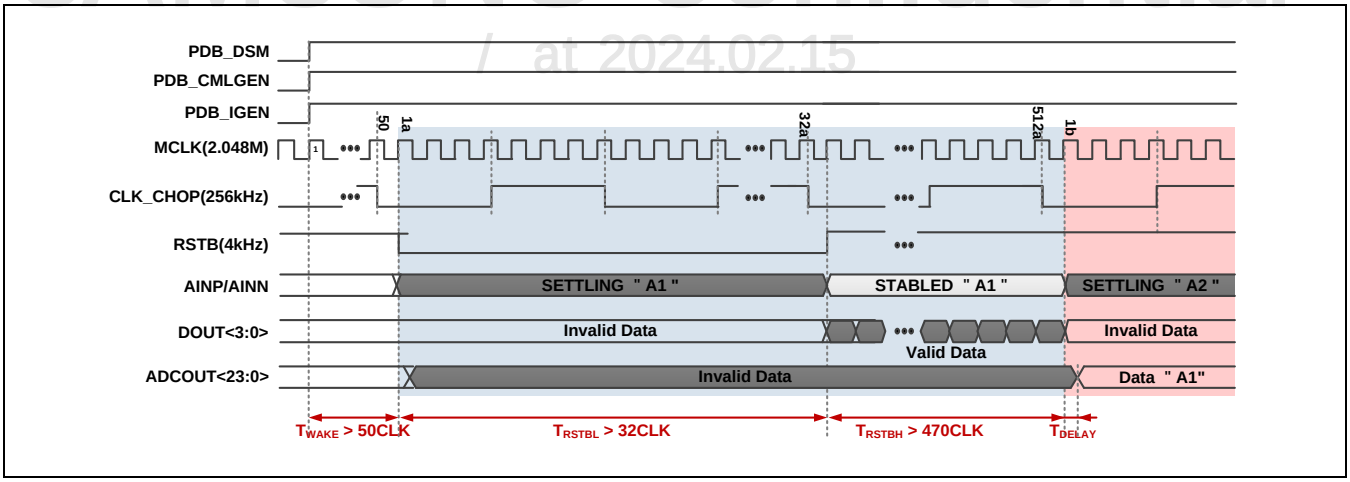


Figure 18 Timing Diagram of Incremental ADC

5.2.2.4.3 Digital Filter for Incremental ADC

A decimation filter is required for ΣΔ modulator output to filter out the out-of-band noise from the modulator. Unlike GSR path, where the ΣΔ modulator is followed by SINC filter (consisting of cascade of integrators and differentiators) and FIR filter, only integrators are employed to form decimation filters in IDC.

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5.2.3 BIA Readout

5.2.3.1 Overview

The BIA applied in S1SBP6A measures the impedance of body to estimate the body fat and body composition. The BIA applies sinusoidal current signals into the body through a pair of electrodes to measure the body composition. BIA readout uses another pair of electrodes to measure voltage signals generated across the impedance level of body. The supply voltage of BIA readout is 1.1 V and the built-in ALDO regulates it. The BIA readout consists of the following components:

- A Direct Digital Synthesizer (DDS) and 6-bit DAC to generate the sinusoidal voltage signal.
- A current driver to apply the generated sinusoidal current signal into the body.
- A built-in input and output switch fabric to select the switches connected to calibration networks or body measuring circuits.
- An input PGA stage to provide additional signal amplification of from 1x to 3x.
- A built-in input HPF to avoid DC offset amplification.
- A clock generation circuits to supply clock signals for Rectify mode.
- An I/Q demodulation circuits to obtain both the real and the imaginary portions of impedance with a single frequency measurement. You can use the same BIA current driver to select BIA readout using either I/Q mode or Rectify mode.
- An output PGA stage to provide additional signal amplification of from 0.5x to 4x.
- A 12-bit ADC for BIA signal digitization. This ADC can be operated at a rate of either 1 kSPS or 8 kSPS with integrated digital decimation filter to provide improved ENOB.

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Figure 19 illustrates the block diagram of BIA readout.

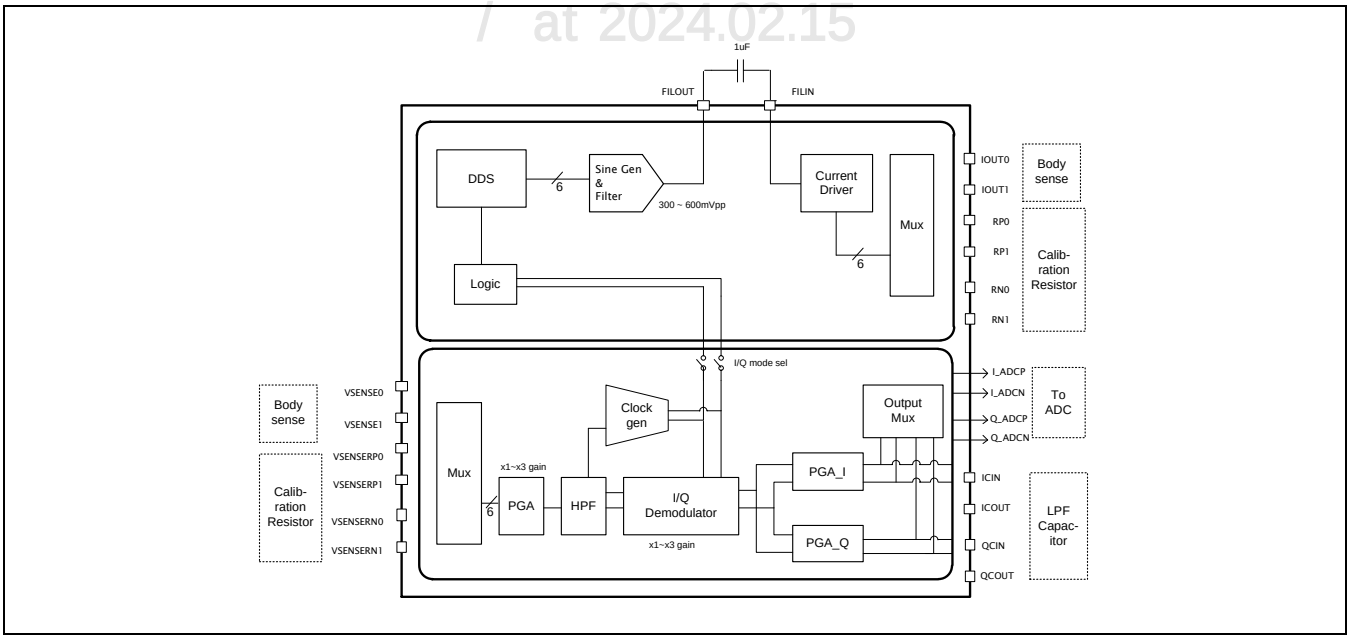


Figure 19 Block Diagram of BIA Readout

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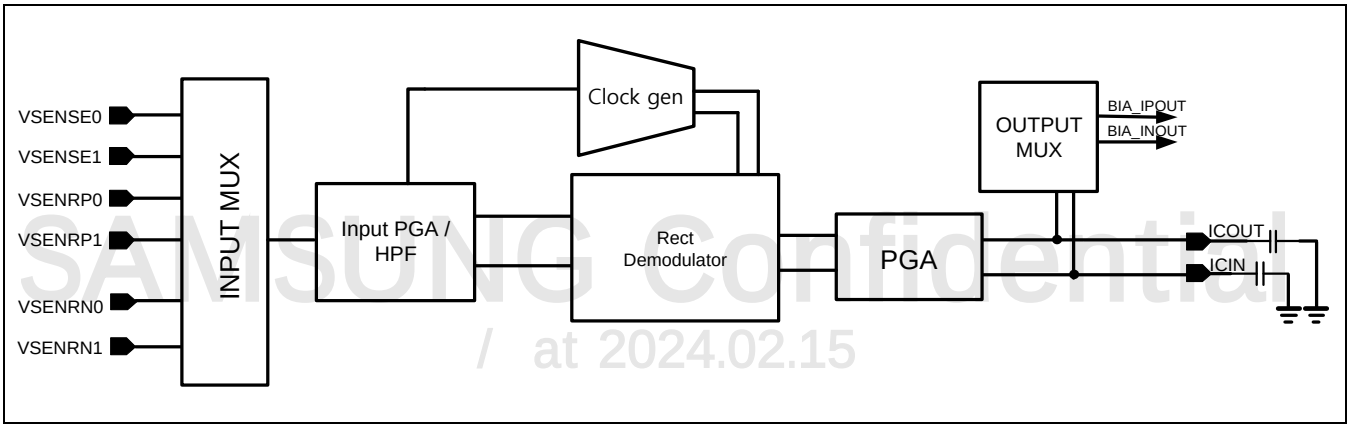
5.2.3.2 Functional Description

You can measure the impedance of the body to calculate the composition and match the result in a table with specified body fat calculation algorithm. The algorithm is usually based on age, sex, weight and so on. The block diagram of BIA readout illustrated in *Figure 19* consists of a sinusoidal current driver as TX and voltage readout as RX. The TX injects current signal whose frequency and the shape of wave are determined by DDS. This current signal modulates impedance of body into voltage signal. Then, RX amplifies the voltage signal at the input PGA and then demodulate the signal using either rectify or I/Q mode.

In the TX, DDS generates 6-bit signals for the voltage DAC to create sinusoidal analog signal which has programmable frequency ranging from 1 kHz to 1 MHz. LPF filters the DAC output. The result is then AC coupled by an external capacitor to restore signal common mode. Finally, an output driver converts it into current signal.

- **Rectify-Mode**

In the Rectify-mode illustrated in *Figure 20*, the readout channel uses self-generated clock signal to perform full wave rectification. The rectified signal is low-pass filtered and used as an input to 12-bit ADC. The LPF uses external large capacitors at its output pins. It is recommended to use one of the several μF capacitors to sufficiently remove the ripples of the demodulation signal.



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Figure 20 Block Diagram of BIA Rectify Mode

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- **I/Q Demodulation Mode**

In the I/Q-mode illustrated in *Figure 21,* BIA RX uses in-phase and quadrature-phase clock signals to demodulate signals and obtain both the real and the imaginary parts of impedance with a single frequency measurement. After it is demodulated, both in-phase and quadrature-phase analog signals are then further amplified by PGAs and low-pass filtered by external large capacitors. The LPF outputs are finally delivered to 12-bit ADC to digitize the real and imaginary values of body impedances. Note that the logical feedback varies the phase of I/Q clock.

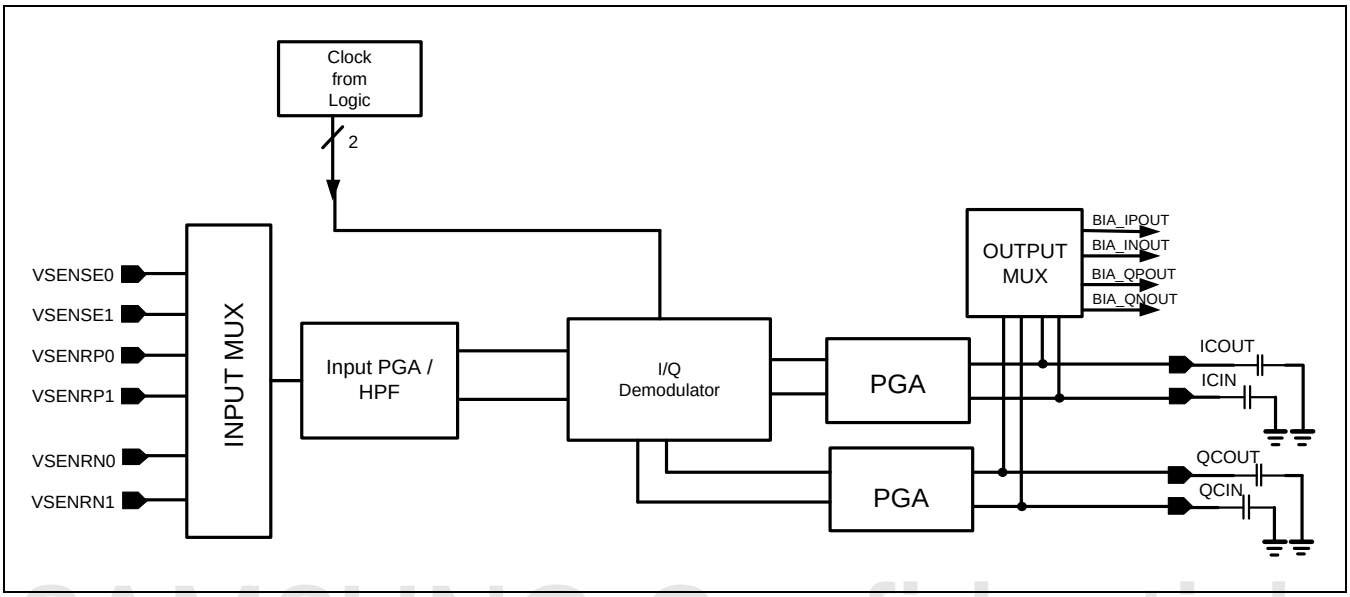


Figure 21 Block Diagram of BIA I/Q Mode

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- **Calibration**

The calibration process is necessary to estimate accurate body impedance. The four external pins (RP0, RP1, RN0, and RN1) are used for the calibration purpose. For example, 300 Ω and 1.5 kΩ resistors are used for the calibration. Firstly, BIA current driver, TX, injects current through 300 Ω and corresponding value is recorded at the RX. Similarly, BIA TX injects same current through 1.5 kΩ resistor and then the corresponding value to 1.5 kΩ is also recorded at the RX. Then, BIA TX injects current to the body. If the impedance value obtained from body is located between 300 Ω and 1.5 kΩ, the practical impedance level of body can be accurately estimated.

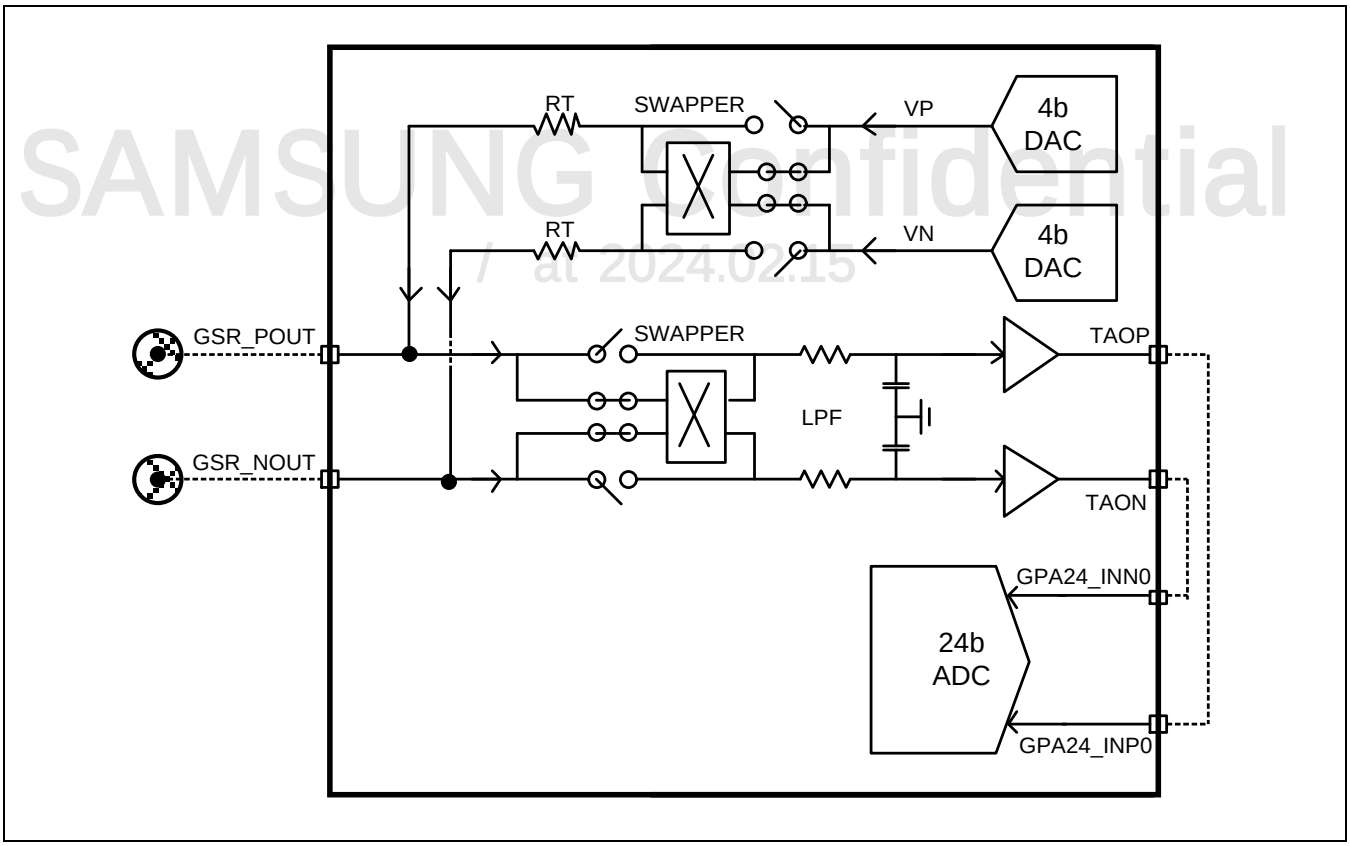
Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

5.2.4 GSR Readout

5.2.4.1 Overview

GSR readout is a low-power integrated AFE dedicated for sensing conductivity of the body skin. As this AFE focuses on measuring conductance, it integrates both voltage sources and readout circuitries. The GSR readout receives supply voltage of 3.0 V from the pin of AVDD33. The GSR readout signal chain illustrated in *Figure 22* consists of the following components:

- GSR_POUT and GSR_NOUT support bipolar electrode montage.
- The GSR readout supports 2.5 V full scale 24-bit ADC. The GSR readout embeds variable resistor. This helps to control the current level that flows into the body through electrodes.
- A built in accurate voltage source provides differential output voltage. The level of the output voltage can be scaled in a wide range to support wide range skin impedance measurement.
- A clock controlled swapper is integrated at the output of voltage source. This feature alternates output voltage signal on the electrodes (AC mode) to prevent electrode polarization. The clock signal can be adjusted to a typical value of 4 Hz. Without enabling clock signal (DC mode), output current level flowing to the body is limited to 4 μA.
- A LPF is integrated at the readout side to limit the noise bandwidth.



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Figure 22 Block Diagram of GSR Readout

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5.2.4.2 Variable Voltage Source

The GSR readout relies on accurate voltage and current generation. In addition, the level of voltage and current signal is variable to support wide range impedance measurement. As illustrated in *Figure 23*, two voltage regulators are integrated to generate accurate voltage signals, VP, and VN. Use the following equations to calculate VP and VN.

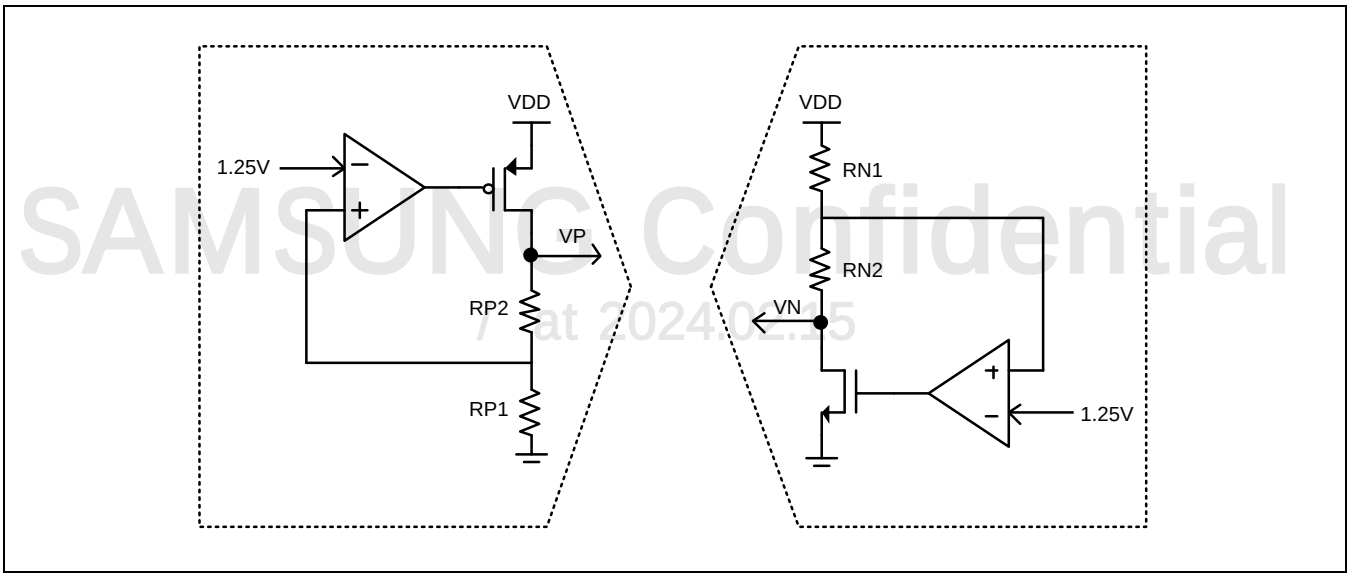
$$\mathbf{VP(V)} = \mathbf{1.25} \times \frac{\mathbf{RP2} + \mathbf{RP1}}{\mathbf{RP1}}$$

$$\mathbf{VN(V)} = \mathbf{VDD} - (\mathbf{VDD} - \mathbf{1.25}) \times \frac{\mathbf{RN2} + \mathbf{RN1}}{\mathbf{RN1}}$$

Following are the sample values to calculate VP and VN.

RN1	RN2	RP1	RP2	VDD
2 MΩ	0.6 –2.1 MΩ	2 MΩ	0.6 –2.1 MΩ	3 V

RN2 and RP2 can be adjusted by the register values of GSR_VNCTRL[3:0] and GSR_VPCTRL[3:0].



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Figure 23 Block Diagram of Voltage Source

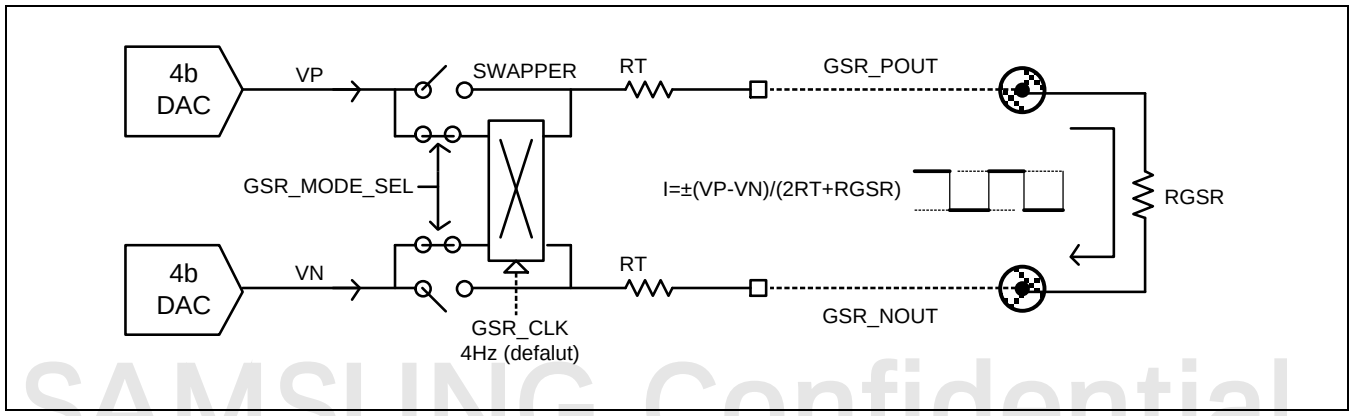
Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

5.2.4.3 Voltage to Current Signal Conversion

As illustrated in *Figure 24*, a swapper is located at the outputs of the voltage source. This swapper is controlled by a clock signal (50 % duty) and outputs alternating voltage signal between VP and VN. A pair of resistors converts the voltage signals into current signals. Use the following equation to calculate the current.

$$\mathbf{I} = \frac{\mathbf{VP} - \mathbf{VN}}{\mathbf{2} \times \mathbf{RT} + \mathbf{RGSR}}$$

Depending on the polarity of the clock signal (AC MODE), the direction of current is swapped between electrodes. The rate of current swapping can be controlled between 0.5 and 16 Hz. Controlling the register value of GSR_EXCTRL[3:0] changes the amount of current which flows through between GSR_POUT and GSR_NOUT. Note that the current direction is fixed by disabling AC MODE (bypass swapper).It is DC MODE.



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Figure 24 Block Diagram of Voltage to Current Signal Conversion

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5.2.4.4 Readout GSR Value

As illustrated in *Figure 25*, the derived current signal flowing into skin impedance, RGSR, generates the voltage signal. This voltage level is converted by an ADC. To eliminate unwanted noise, a LPF is inserted with its programmable bandwidth. The bandwidth is adjusted by controlling the register value of GSR_LPCTRL[3:0] to adjust the value of RLPF. Similar to current generation, the swapper at the readout is also configured between AC MODE and DC MODE by the same register value described above. RGSR can be calculated using the following formula.

$$\mathbf{RGSR} = \frac{\mathbf{2} \times \mathbf{RT} \times \left(\mathbf{ADCcode} \times \frac{\mathbf{VREF}}{\mathbf{2}^{N-2}}\right)}{(\mathbf{VP} - \mathbf{VN}) - \left(\mathbf{ADCcode} \times \frac{\mathbf{VREF}}{\mathbf{2}^{N-2}}\right)}$$

Where N is 24-bit ADC resolution and VREF is reference voltage (2.5 V) for ADC.
To remove variations of RT in each sample, the exact RT value can be calculated as follows.

$$\mathbf{RT_{real}} = \frac{\mathbf{10M\Omega} \times (\mathbf{Vsource_code} - \mathbf{ADCcode@10M\Omega})}{\mathbf{2} \times (\mathbf{ADCcode@10M\Omega} - \mathbf{offset_code})},$$

$$\mathbf{Vsource_code} = (\mathbf{VP} - \mathbf{VN}) \times \left(\frac{\mathbf{2}^{N-2}}{\mathbf{VREF}}\right)$$

The value of ADCcode measured at 10 MΩ is stored in each sample, and the values of both Vsource_code and offset_code are also stored for more accurate calculation. As a result, the exact RGSR can be calculated using the following simple formula.

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$$\mathbf{RGSR} = \frac{\mathbf{2} \times \mathbf{RT_{real}} \times (\mathbf{ADCcode} - \mathbf{offset_code})}{\mathbf{Vsource_code} - \mathbf{ADCcode}}$$

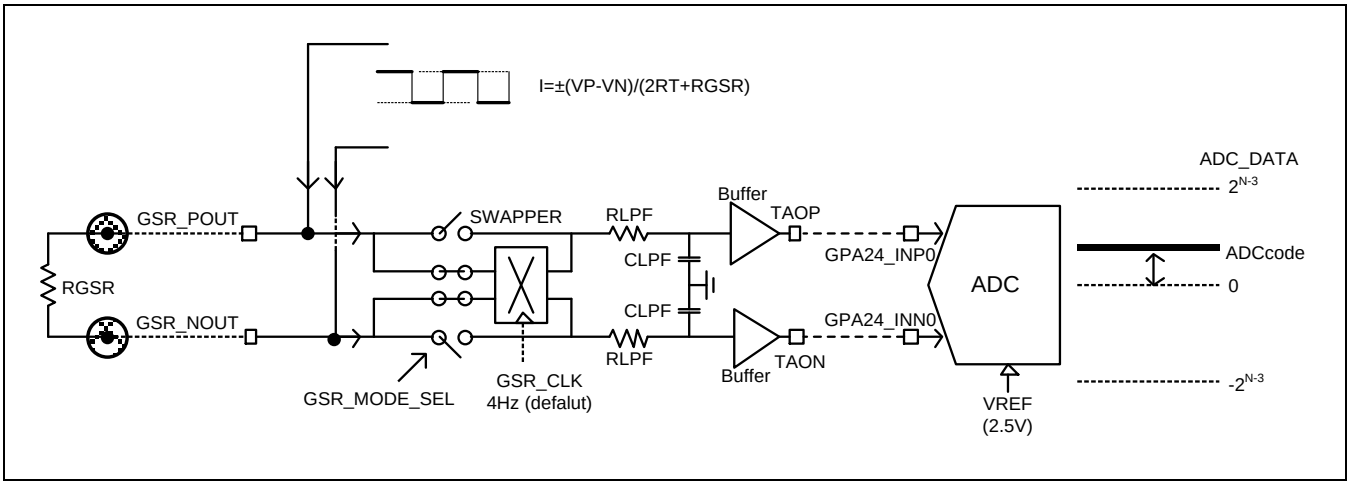


Figure 25 Block Diagram of GSR Readout with LPF

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5.2.4.5 ADC Timing Diagram

ADC samples the GSR output signals and converts it into 24-bit data stream (ADC_DATA in *Figure 26*).

For high-resolution GSR readout, sampling frequency of ADC (ADC_SMPL in *Figure 26*) is 64 kHz and the oversampling ratio (OSR) is configured to 64, 128, 256, or 512. It is recommended to reset the ADC at start by the register value of SD_RST_MOD. When OSR is 128 (500 SPS), there is delay of 64 ms to output meaningful GSR data. *Figure 26* illustrates the timing diagram of high-resolution GSR readout.

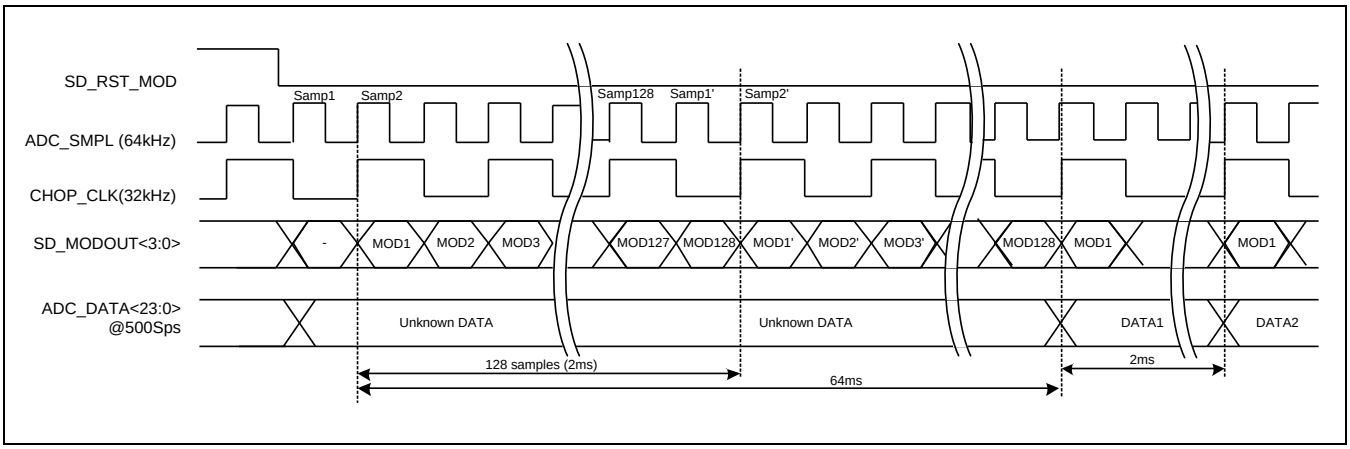


Figure 26 Timing Diagram of High-Resolution GSR Readout

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5.2.5 GP Analog Interfaces

5.2.5.1 12-bit Analog Readout (GP Analog0/1)

12-bit analog readout front-end, as illustrated in *Figure 27*, is integrated to support versatile sensor interface applications where low-power and medium-range resolution is the primary concern. Prior to ADC, highly flexible AFEs are embedded. They are grouped into two in which General Purpose (GP) Analog0 functions as a Variable Gain Amplifier (VGA), a Trans-Impedance Amplifier (TIA), or a LPF, and GP Analog1 behaves as a voltage attenuator and a die temperature sensor. The flexibilities of GP Analog 0/1 are realized by integrated switch matrix and you can easily program them based on the preference. The outputs of GP Analog 0/1 are digitized by a 12-bit ADC operating at a rate of 1 kSPS or 8 kSPS in a time-multiplexed manner.

- LPF in GP Analog0 provides first-order differential mode bandwidth from 1 kHz to 16.3 kHz.
- Voltage attenuator supports input voltage signals up to 1.4 V_{p-p} with attenuation of 1/5 (For example, Rext = 200 kΩ). Note that resistor of 100 kΩ is internally integrated.
- ADC for GP Analog0/1 operates at a rate of either 1 kSPS or 8 kSPS and it outputs 12-bit digital code.

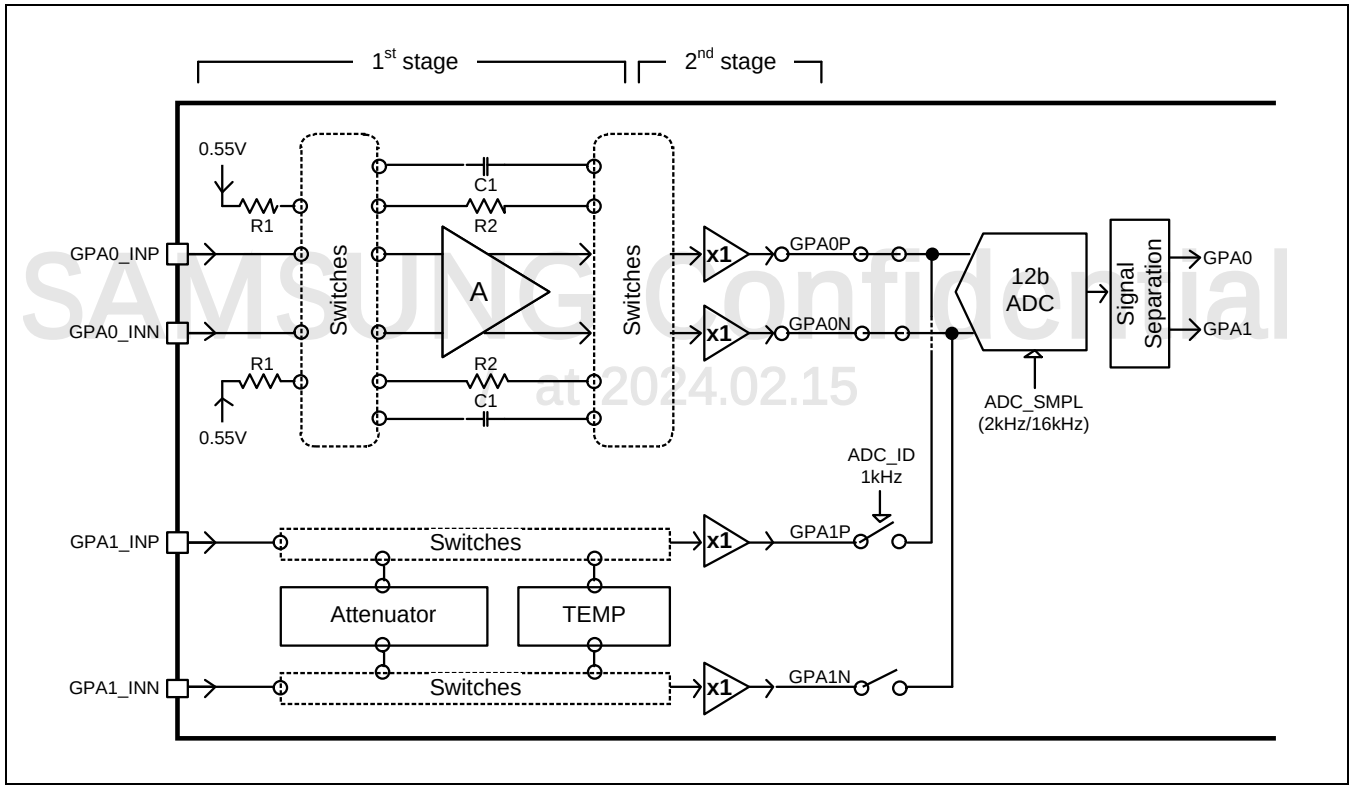


Figure 27 Block Diagram of 12 bit Analog Readout

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

5.2.5.1.1 Switch Configurations for GP Analog0

GP Analog0 Integrates 12 switches which can be configured by the register value of GPA0_SW_SEL[11:0]. *Table 19* describes the detail switch configurations in GP Analog0.

Table 19 Switch Configurations in GP Analog0

Stage	Interface	GPA0_SW_SEL[11:0]											
		11	10	9	8	7	6	5	4	3	2	1	0
1st stage	TIA	–	–	0	0	1	1	1	1	0	1	1	0
	VGA	–	–	0	0	1	1	1	1	1	0	0	1
	LPF	–	–	0	1	1	0	0	1	0	1	1	0
	Bypass (1st-stage)	–	–	1	0	0	0	0	0	0	0	0	0
2nd stage	ADC Driver	1	0	–	–	–	–	–	–	–	–	–	–
	Bypass (2nd-stage)	0	1	–	–	–	–	–	–	–	–	–	–

5.2.5.1.1.1 TIA in GP Analog0

For TIA configuration illustrated in *Figure 28*, GPA0_INP/INN can receive differential current signal (for example, from Photo Diode). The TIA provides differential trans-impedance gain of 2 x R2. The value of R2 can be adjusted in five steps according to the register value of GPA0_R2_CTRL[1:0] and GPA0_R2_300K. The TIA also equips capacitor C1 in parallel with R2. This arrangement can provide first-order anti-alias filter for the ADC. In addition, C1 helps to compensate for the stability when there is pronounced capacitance presented at inputs. The value of C1 can be configured in four steps according to the register value of GPA0_C1_CTRL[1:0], and it provides wide range of -3 dB bandwidth depending on the value of R2. Internally, the TIA output is biased to the half of ADC reference voltage, 0.55 V. Use the following equations to determine the gain and bandwidth of TIA.

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$$\text{Gain (V/A)} = 2 \cdot R_2$$

$$\text{Bandwidth (Hz)} = \frac{1}{2 \cdot \pi \cdot R_2 \cdot C_1}$$

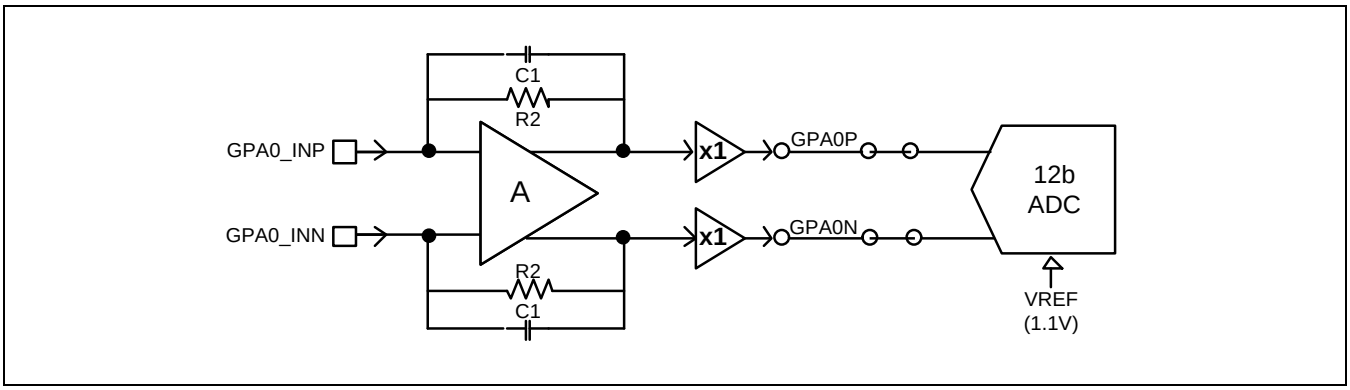


Figure 28 Block Diagram of TIA Configuration in GP Analog0

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Table 20 describes the detail configurations for the gain and bandwidth of TIA in GP Analog0.

Table 20 Gain and Bandwidth of TIA in GP Analog0

GPA0_R2_300K	GAP0_R2_CTRL[1:0]	GPA0_C1_CTRL[1:0]	Gain [MΩ]	–3 dB BW [kHz]
0	00	00	1.30	16.29
		01		12.22
		10		9.77
		11		8.14
	01	00	2.60	8.14
		01		6.11
		10		4.89
		11		4.07
	10	00	5.21	4.07
		01		3.05
		10		2.44
		11		2.04
	11	00	10.42	2.04
		01		1.53
		10		1.22
		11		1.02
1	XX	00	0.60	35.29
		01		26.47
		10		21.17
		11		17.65

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5.2.5.1.1.2 VGA in GP Analog0

For VGA configuration illustrated in *Figure 29*, GPA0_INP/INN can receive differential voltage signal (for example, from bridge circuit). The VGA provides pseudo-differential voltage gain of 1+R2/R1. This gain can be adjusted in four steps according to the register value of GPA0_R1_CTRL[1:0]. Note that the value of R2 for VGA is cannot be adjusted. Similar to TIA, C1 in parallel with R2 provides first-order anti-alias filter for the ADC. The value of C1 can be configured in four steps according to the register value of GPA0_C1_CTRL[1:0]. Note that the common-mode voltage to GPA0_INP/INN must be 0.55 V as the VGA behaves as a non-inverting amplifier and internally biased. Use the following equations to determine the gain and bandwidth of VGA.

$$\mathbf{Gain\ (V/V)} = \mathbf{1} + \frac{\mathbf{R_2}}{\mathbf{R_1}}$$

$$\mathbf{Bandwidth\ (Hz)} = \frac{\mathbf{1}}{\mathbf{2 \cdot \pi \cdot R_2 \cdot C_1}}$$

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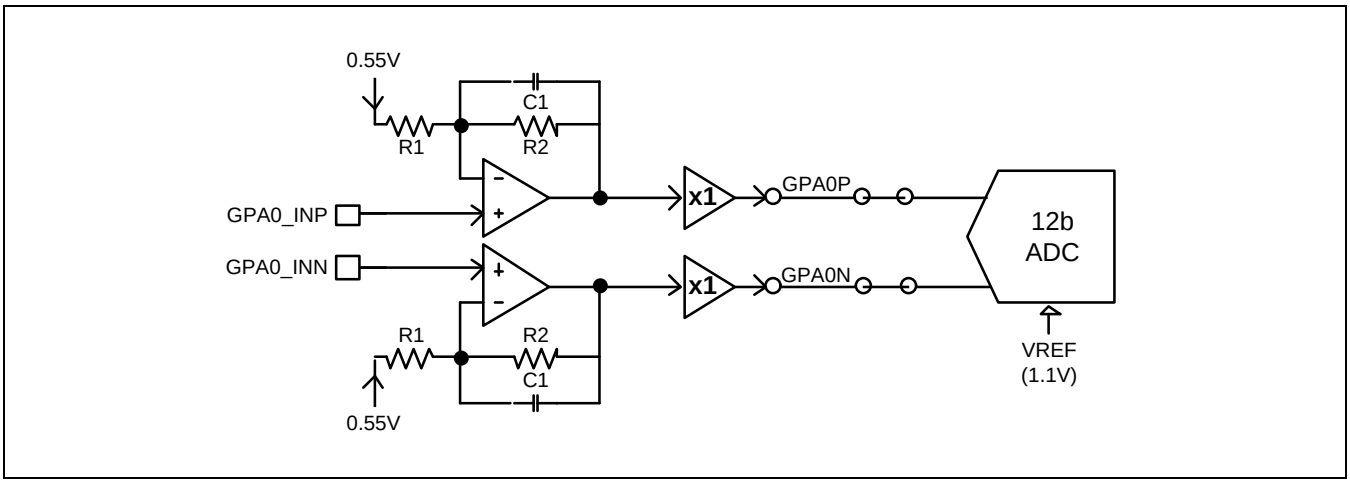


Figure 29 Block Diagram of VGA Configuration in GP Analog0

Table 21 describes the detail configurations for the gain and bandwidth of VGA in GP Analog0.

Table 21 Gain and Bandwidth of VGA in GP Analog0

GPA0_R2_CTRL[1:0]	GAP0_R1_CTRL[1:0]	Gain [V/V]
11	00	15.84
	01	7.94
	10	3.98
	11	1.99

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GPA0_R2_CTRL[1:0]	GAP0_C1_CTRL[1:0]	BW [kHz]
11	00	2.04
	01	1.53
	10	1.22
	11	1.02

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5.2.5.1.1.3 LPF in GP Analog0

For LPF configuration illustrated in *Figure 30*, R2 and C1 provide passive LPF prior to ADC. In this configuration, all amplifiers are turned off for low-power consumption. Unlike VGA, the common-mode voltage inputs at GPA0_INP/INN are not limited to 0.55 V. Depending on the register value of R2 and C1, the –3 dB bandwidth is configured in wide range from 1 kHz to 16.3 kHz. Use the following equations to determine the bandwidth of LPF.

$$\textbf{Bandwidth (Hz)} = \frac{1}{2 \cdot \pi \cdot R_2 \cdot C_1}$$

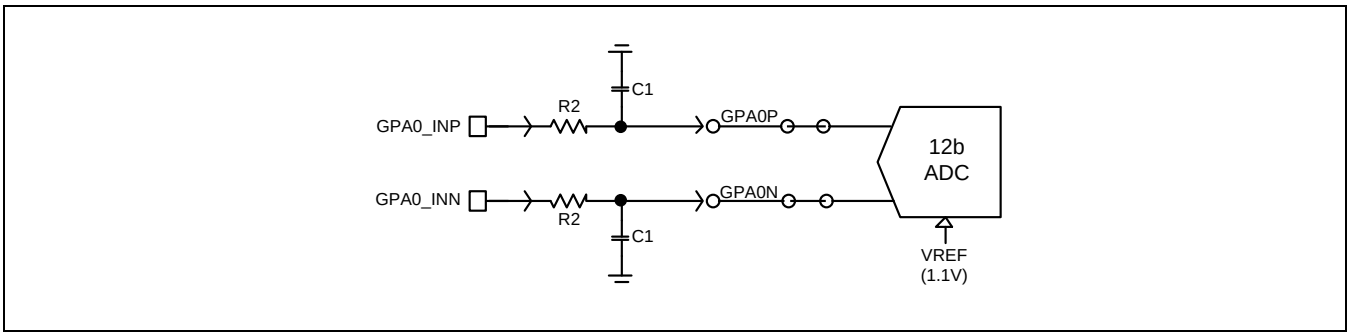


Figure 30 Block Diagram of LPF Configuration in GP Analog0

Table 22 describes the detail configurations for bandwidth of LPF in GP Analog0.

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Table 22 Bandwidth of LPF in GP Analog0

GPA0_R2_CTRL[1:0]	GAP0_C1_CTRL[1:0]	BW [kHz]
00	00	16.29
	01	12.22
	10	9.77
	11	8.14
01	00	8.14
	01	6.11
	10	4.89
	11	4.07
10	00	4.07
	01	3.05
	10	2.44
	11	2.04
11	00	2.04
	01	1.53
	10	1.22
	11	1.02

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5.2.5.1.2 Switch Configurations for GP Analog1

GP Analog1 integrates nine switches which can be configured by the register value of GPA1_SW_SEL[8:0]. *Table 23* describes the detail switch configurations.

Table 23 Switch Configurations in GP Analog1

Stage	Interface	GPA1_SW_SEL[8:0]								
		8	7	6	5	4	3	2	1	0
1st stage	Attenuator	0	–	–	0	1	0	1	1	0
	Die temperature	0	–	–	1	1	0	0	0	0
	Bypass (1st-stage)	0	–	–	0	1	0	0	1	0
	GP Switch	0	–	–	0	1	0	0	0	1
2nd stage	ADC driver	0	1	0	–	–	–	–	–	–
	Bypass (2nd-stage)	0	0	1	–	–	–	–	–	–

5.2.5.1.2.1 Voltage Attenuator in GP Analog1

GP Analog1 integrates 100 kΩ internal resistor, Rint, at inputs (between GPA1_INP and GPA1_INN). With the help of external resistor in series, Rext illustrated in *Figure 31*, it can behave as a voltage attenuator. Use the following equation to determine the voltage attenuation configuration in GP analog1.

$$\text{Attenuation (V/V)} = \frac{\text{Rint}}{2 \cdot \text{Rext} + \text{Rint}}$$

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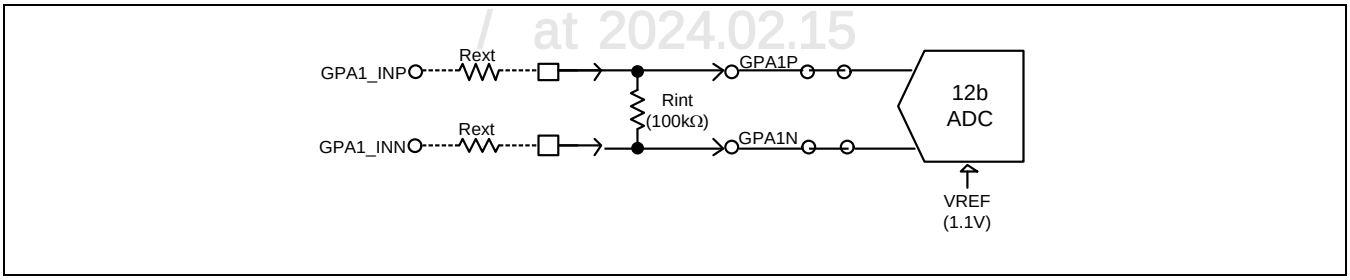


Figure 31 Block Diagram of Voltage Attenuator Configuration in GP Analog1

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5.2.5.1.2.2 Die Temperature Sensor in GP Analog1

GP Analog1 integrates thermal diodes as illustrated in *Figure 32*. The ADC can measure voltage difference of diode junctions with respect to temperature. Use the following equation to determine the voltage difference of diode junctions and ADC code.

$$\mathbf{V1 - V2\ (V)} = \frac{\mathbf{kT}}{\mathbf{q}} \times \mathbf{ln27}$$

$$\mathbf{ADC\ code} = \mathbf{2048} + \frac{\mathbf{V1 - V2}}{\mathbf{1LSB}}$$

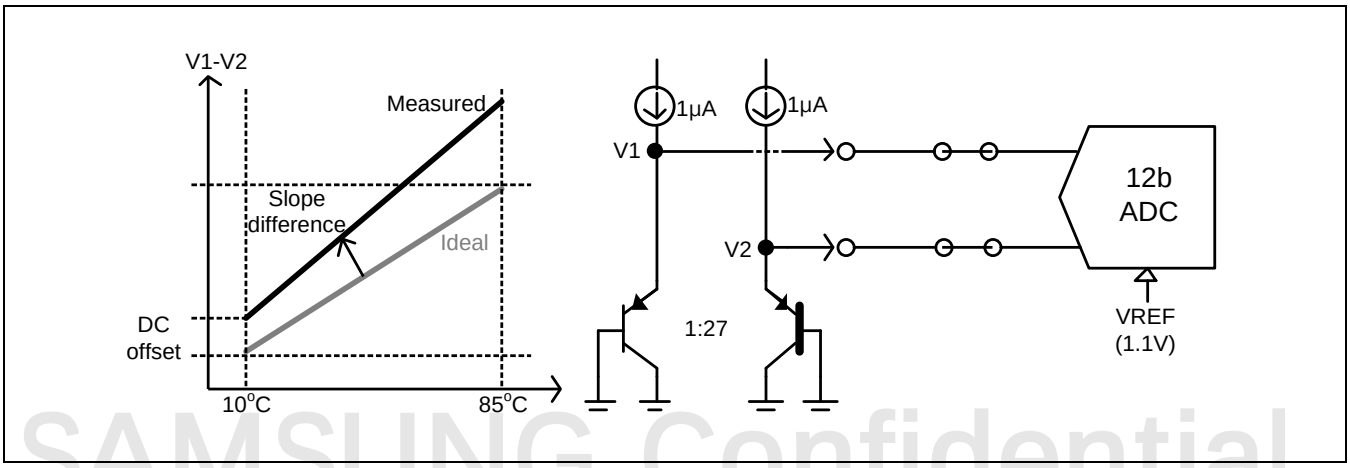


Figure 32 Block Diagram of Die Temperature Sensor Configuration in GP Analog1

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Practically, there exists slope difference and DC offset between measured transfer curve and an ideal transfer curve.

To compensate this, V1 and V2 are measured at temperature of 10 °C and 85 °C, respectively. The measured 12 bit ADC code is then compensated (DC offset and slope difference) by the digital signal processing (Software). The following equation is considered for the transfer curve compensation to determine the calibrated temperature value of Tsense.

$$\mathbf{Tcalib} = (\mathbf{Tsense} - \mathbf{TE1}) \times \frac{\mathbf{D85 - D10}}{\mathbf{TE2 - TE1}} + \mathbf{TE1} - (\mathbf{TE1 - D10})$$

Where Tsense is the practically measured ADC code including DC offset and slope error. Tcalib is the calibrated ADC code of Tsense, TE1 corresponds to the measured ADC code at 10 °C, and TE2 corresponds to the measured ADC code at 85 °C. D85 is the ideal ADC code(=2237) at 85 °C, and D10 is the ideal ADC code(=2198) at 10 °C. According to the above equation, the last term (TE1- D10) represents DC offset and the remaining term represents the slope compensation.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

5.2.5.1.2.3 Analog Switch in GP Analog1

GP Analog1 provides single analog switch illustrated in *Figure 33* which can be programmed by register value of GPA1_SW_SEL[0]. GPA1_INP and GPA1_INN are used as switch terminals. Note that other circuits such as temperature sensor and ADC are completely isolated from GPA1_INP/INN in this configuration.

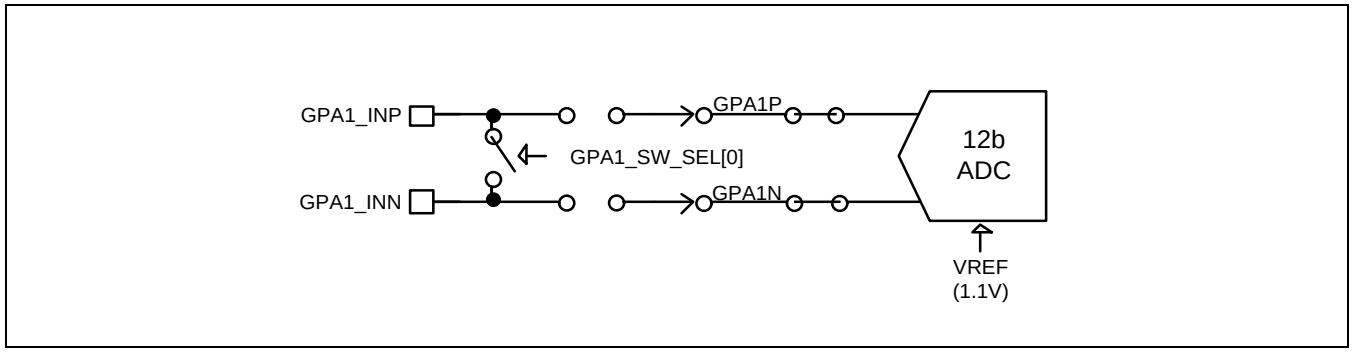


Figure 33 Block Diagram of Analog Switch in GP Analog1

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5.2.6 FIFO Control

5.2.6.1 Overview

The AFE control logic has the following four functions:

- SAR-ADC channel separation/decimation
- Six 16-bit FIFOs and three 24-bit FIFO
- DMA controller of FIFO
- Special function registers for AFE controller

5.2.6.2 Feature

AFE Controller has nine FIFOs, six FIFOs for 12 bits SAR-ADCs, one FIFO for 24 bits Sigma-Delta ADC, and two FIFOs for 24 bits Incremental ADC. Each FIFO has 16 depths. All FIFOs operate in PCLK domain. Each FIFO has three interrupt sources which are FULL, EMPTY, and READY. Only one combined interrupt signal is generated from FIFOs and it can be cleared by writing 1 to INTR_CLR register. Interrupt masking is available using FIFO_INTR_MASK registers. Ready interrupt is generated when the number of data saved in FIFO is larger than the number of data set by FIFO_READY_SIZE_SET register. Register control also supports software reset of FIFO. DMA controller in FIFO generates DMA request signals according to the register setting. Use FIFO_READ_DATA_SIZE_SET register to set the number of data which generates the DMA signal. After the controller gets acknowledge signal from µDMA, the request signals are cleared automatically. *Figure 34* illustrates the block diagram of AFE FIFO controller.

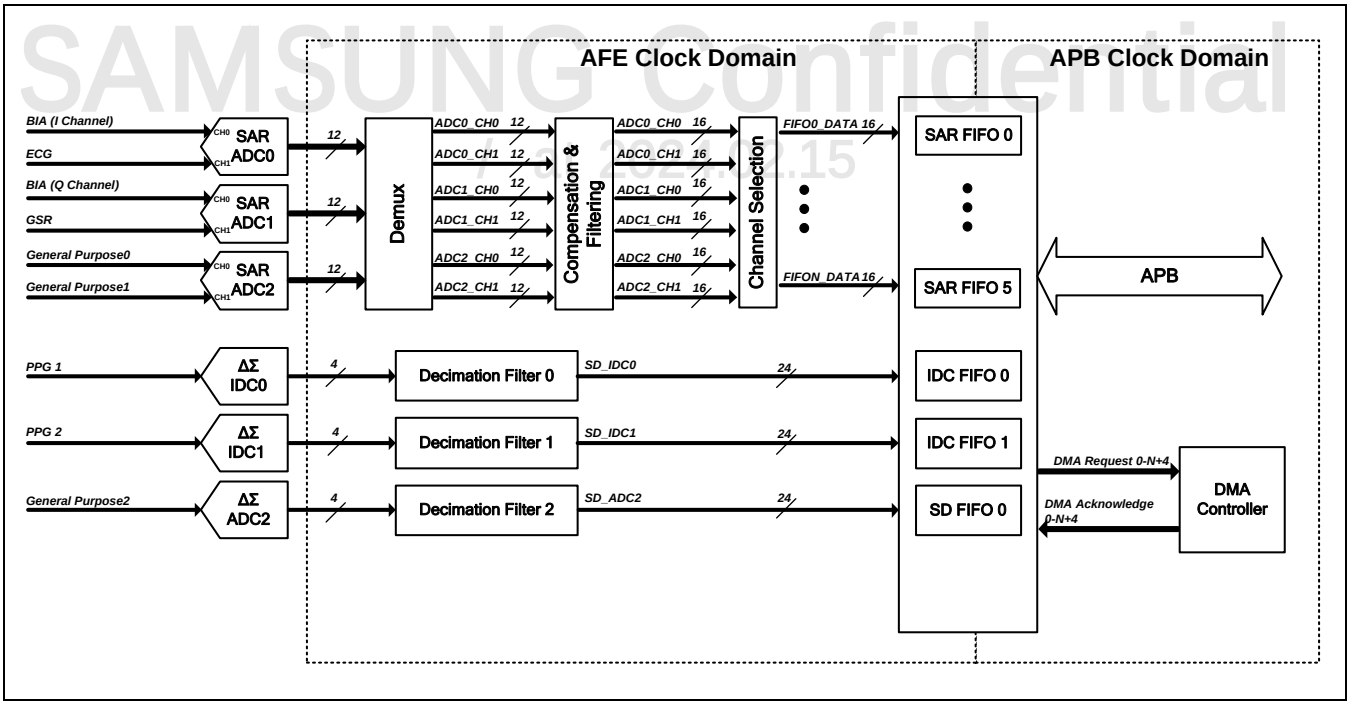
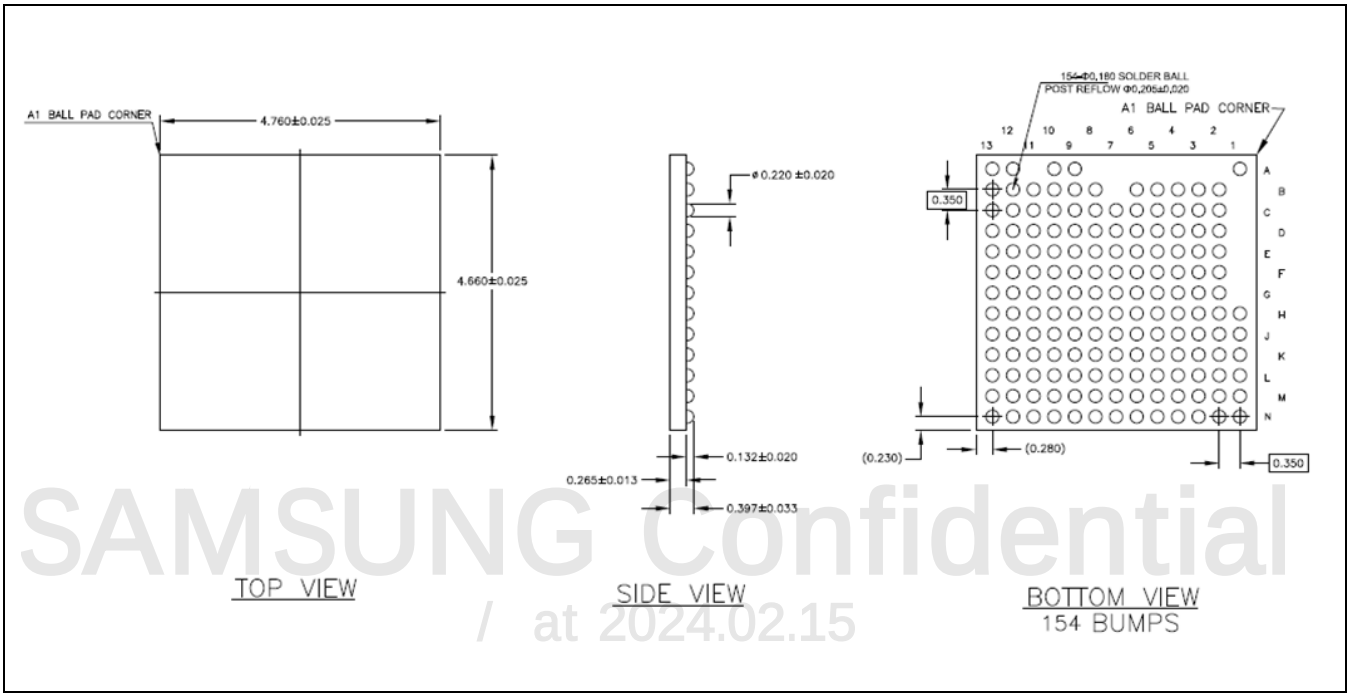


Figure 34 Block Diagram of AFE FIFO Controller

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6 Mechanical Data

Figure 35 illustrates the 154 pin WLP outline.



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Figure 35 154 Pin WLP (4.76 × 4.66 × 0.43 mm) Outline

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